Unit 8. Generalised Linear Models II: Counts and the Exponential Family

1 Counts and the Poisson Distribution

FREES §12.1.1; ISLR §4.6.1, §4.6.2 (PA reading list, not SRM). SRM learning outcomes 2(a), 2(b).

Remark (The classical assumptions; Unit 2)**.** A1–A5 are Unit 2's classical assumptions on $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}$ fixed: linearity (A1), zero-mean errors (A2), $\text{var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$ with $\sigma^2 > 0$ (A3), $\text{rank}(\mathbf{X}) = k+1$, with $n > k+1$ for inference (A4), and normal errors (A5).

1.1 The Poisson distribution

Definition 8.1 (Poisson; FREES §12.1.1, ISLR §4.6.2)**.** A **count** takes values in $\{0, 1, 2, \ldots\}$. A count y has the **Poisson** distribution with mean $\mu > 0$, written $y \sim \text{Pois}(\mu)$, if

$$\mathsf{P}(y = j) = \frac{\mu^j}{j!}\, e^{-\mu}, \qquad j = 0, 1, 2, \ldots \tag{1}$$

Proposition 8.2 (Moments and additivity; FREES §12.1.1)**.** Let $y \sim \text{Pois}(\mu)$.

(i) $\mathsf{E}[s^y] = e^{\mu(s-1)}$ for every $s \in \mathbb{R}$.

(ii) $\mathsf{E}\, y = \mu$ and $\operatorname{var} y = \mu$ (**equidispersion**); every moment is finite, and $\mathsf{E}(y-\mu)^3 = \mu$, $\mathsf{E}(y-\mu)^4 = \mu + 3\mu^2$.

(iii) If $y_1, \dots, y_m$ are independent with $y_j \sim \text{Pois}(\mu_j)$, then $\sum_j y_j \sim \text{Pois}(\sum_j \mu_j)$.

Proof. (i) $\mathsf{E}[s^y] = e^{-\mu}\sum_{j\geqslant 0}(\mu s)^j/j! = e^{-\mu}e^{\mu s}$, the series converging absolutely for every s.
(ii) For $r \geqslant 1$, $\mathsf{E}[y(y-1)\cdots(y-r+1)] = e^{-\mu}\sum_{j\geqslant r}\mu^j/(j-r)! = \mu^r$, the terms with $j < r$ vanishing; hence every moment is a finite combination of these and is finite. With $r = 1, 2, 3, 4$: $\mathsf{E}\, y = \mu$, $\mathsf{E}\, y^2 = \mu^2+\mu$, $\mathsf{E}\, y^3 = \mu^3+3\mu^2+\mu$ and $\mathsf{E}\, y^4 = \mu^4+6\mu^3+7\mu^2+\mu$, using $y^2 = y(y-1)+y$, $y^3 = y(y-1)(y-2)+3y(y-1)+y$ and $y^4 = y(y-1)(y-2)(y-3)+6y(y-1)(y-2)+7y(y-1)+y$. Expanding the central moments binomially, $\operatorname{var} y = \mu$, $\mathsf{E}(y-\mu)^3 = \mathsf{E}\, y^3 - 3\mu\,\mathsf{E}\, y^2 + 2\mu^3 = \mu$ and $\mathsf{E}(y-\mu)^4 = \mathsf{E}\, y^4 - 4\mu\,\mathsf{E}\, y^3 + 6\mu^2\,\mathsf{E}\, y^2 - 3\mu^4 = \mu + 3\mu^2$.
(iii) By independence the generating function of the sum is $\prod_j e^{\mu_j(s-1)} = e^{(\sum_j \mu_j)(s-1)}$, the generating function of $\text{Pois}(\sum_j \mu_j)$. The law itself follows by direct convolution: for two independent summands, $\mathsf{P}(y_1+y_2 = t) = \sum_{a=0}^{t} e^{-\mu_1}\frac{\mu_1^a}{a!}e^{-\mu_2}\frac{\mu_2^{t-a}}{(t-a)!} = \frac{e^{-(\mu_1+\mu_2)}}{t!}\sum_{a=0}^{t}\binom{t}{a}\mu_1^a\mu_2^{t-a} = e^{-(\mu_1+\mu_2)}\frac{(\mu_1+\mu_2)^t}{t!}$ by the binomial theorem, and induction on the number of summands gives the general case. □

Proposition 8.3 (The sample mean is the maximum likelihood estimate; FREES §12.1.1)**.** Let $y_1, \ldots, y_n$ be independent Pois(μ) with $\bar{y} > 0$. The log-likelihood

$$L(\mu) = \sum_{i=1}^{n} \left(-\mu + y_i \ln \mu - \ln y_i!\right), \qquad \mu > 0, \tag{2}$$

has the unique maximiser $\hat{\mu} = \bar{y}$. If $\bar{y} = 0$ it has no maximiser on $(0, \infty)$.

Proof. $L'(\mu) = -n + n\bar{y}/\mu$ and $L''(\mu) = -n\bar{y}/\mu^2 < 0$ when $\bar{y} > 0$, so L is strictly concave (the prerequisites' lemma **convexity from the derivative**) with its only stationary point at $\bar{y}$, its unique maximiser by the prerequisites' proposition **first-order condition**. If $\bar{y} = 0$ then $L(\mu) = -n\mu$ is strictly decreasing on $(0, \infty)$. □

Example 8.1 (Singapore automobile claims; FREES §12.1.1, Table 12.1)**.** FREES's $n = 7483$ automobile policies of a Singapore insurer in 1993, with `Clm_Count` the number of accidents in the year:

```
1 S <- read.csv("data/frees/csv/SingaporeAuto.csv", quote = "")
```

```
y <- S$Clm_Count; n <- length(y)
table(y); ybar <- mean(y); ybar
```

```
## y
##    0    1    2    3
## 6996  455   28    4
## [1] 0.06989
```

```
round(n * dpois(0:4, ybar), 2)
```

```
## [1] 6977.86  487.69   17.04    0.40    0.01
```

The mean is $\bar{y} = 0.06989$, and by Proposition 8.3 the fitted counts $n\,\mathsf{P}(y = j)$ at $\hat{\mu} = \bar{y}$ are 6977.86, 487.69 (487.6948 to four decimals; FREES prints 487.70), 17.04, 0.40 and 0.01, FREES's Table 12.1.

1.2 Pearson's goodness-of-fit statistic

Definition 8.4 (Pearson's chi-square statistic; FREES (12.3))**.** Let $C_1, \dots, C_c$ ($c \geqslant 2$) partition $\{0, 1, 2, \dots\}$ into **cells**, let $N_j = \#\{i : y_i \in C_j\}$ be the observed and $n\hat{p}_j$ the fitted count of cell j. The **Pearson statistic** is

$$X^2 = \sum_{j=1}^{c} \frac{(N_j - n\hat{p}_j)^2}{n\hat{p}_j}.$$

Theorem 8.5 (Pearson's statistic with known cell probabilities)**.** Let $y_1, \dots, y_n$ be independent and identically distributed with $p_j = \mathsf{P}(y_1 \in C_j) > 0$ for every j, and take $\hat{p}_j = p_j$. Then $X^2 \xrightarrow{d} \chi^2_{c-1}$.

Proof. Put $\mathbf{z}_i = (\mathbb{1}\{y_i \in C_1\} - p_1, \dots, \mathbb{1}\{y_i \in C_{c-1}\} - p_{c-1})'$, the first $c-1$ cells only. The $\mathbf{z}_i$ are independent and identically distributed, bounded by 1 in each coordinate, with mean $\mathbf{0}$ and covariance $\boldsymbol{\Sigma} = \mathbf{D} - \mathbf{p}\mathbf{p}'$, where $\mathbf{p} = (p_1, \dots, p_{c-1})'$ and $\mathbf{D} = \operatorname{diag}(\mathbf{p})$, since $\mathbb{1}\{y \in C_j\}\,\mathbb{1}\{y \in C_l\} = \mathbb{1}\{y \in C_j\}\delta_{jl}$. By the prerequisites' lemma **Sherman–Morrison**, with $\mathbf{1}$ the vector of

ones and $1 - \mathbf{p}'\mathbf{D}^{-1}\mathbf{p} = 1 - \sum_{j<c} p_j = p_c > 0$,

$$\boldsymbol{\Sigma}^{-1} = \mathbf{D}^{-1} + \frac{\mathbf{D}^{-1}\mathbf{p}\mathbf{p}'\mathbf{D}^{-1}}{1 - \mathbf{p}'\mathbf{D}^{-1}\mathbf{p}} = \mathbf{D}^{-1} + \frac{\mathbf{1}\mathbf{1}'}{p_c},$$

so $\boldsymbol{\Sigma}$ is positive definite. The Lyapunov corollary of the prerequisites, with $\delta = 1$, gives $\mathbf{V}_n = n^{-1/2}\sum_i \mathbf{z}_i \xrightarrow{d} N(\mathbf{0}, \boldsymbol{\Sigma})$. Coordinate j of $\mathbf{V}_n$ is $(N_j - np_j)/\sqrt{n}$, and $\sum_{j\leqslant c}(N_j - np_j) = n - n = 0$, so $\mathbf{1}'\mathbf{V}_n = -(N_c - np_c)/\sqrt{n}$. Hence

$$\mathbf{V}_n'\boldsymbol{\Sigma}^{-1}\mathbf{V}_n = \sum_{j<c} \frac{(N_j - np_j)^2}{np_j} + \frac{(N_c - np_c)^2}{np_c} = X^2$$

exactly, and the prerequisites' theorem **quadratic forms of asymptotically normal vectors**, with $\hat{\boldsymbol{\Sigma}}_n = \boldsymbol{\Sigma}$, gives the χ^2_{c-1} limit. □

Lemma 8.6 (Cell probabilities of the Poisson)**.** For $y \sim \text{Pois}(\mu)$ and cells $C_1, \ldots, C_c$, let $p_j(\mu) = \mathsf{P}(y \in C_j)$ and $m_j(\mu) = \mathsf{E}[y \mid y \in C_j]$ when $p_j > 0$. Then each p_j is differentiable on $(0, \infty)$ with

continuous derivative

$$p_j'(\mu) = \frac{\mathsf{E}[\mathbb{1}\{y \in C_j\}(y-\mu)]}{\mu} = \frac{p_j(m_j-\mu)}{\mu}, \qquad \sum_{j=1}^{c} p_j'(\mu) = 0,$$

and

$$\kappa(\mu) = \mu \sum_{j=1}^{c} \frac{p_j'(\mu)^2}{p_j(\mu)} = \frac{\sum_j p_j(m_j-\mu)^2}{\mu} \in [0,1).$$

Proof. $p_j(\mu) = \sum_{y\in C_j} g(y,\mu)$ with $g(y,\mu) = e^{-\mu}\mu^y/y!$, a sum against counting measure, and $\partial g/\partial\mu = g\cdot(y/\mu - 1)$. For μ in $[\mu_0/2, 2\mu_0]$, $|\partial g/\partial\mu| \leqslant e^{-\mu_0/2}(2\mu_0)^y(2y/\mu_0+1)/y!$, which is summable over y; the prerequisites' theorem **differentiation under the integral sign** gives $p_j'(\mu) = \sum_{y\in C_j} g(y,\mu)(y-\mu)/\mu = \mathsf{E}[\mathbb{1}\{y\in C_j\}(y-\mu)]/\mu$. The same argument applied to this series, whose terms are g times a polynomial in y and $1/\mu$, shows p_j' is itself differentiable, hence continuous. Summing over j gives $\mathsf{E}(y-\mu)/\mu = 0$. Since $\mathsf{E}[\mathbb{1}\{y\in C_j\}y] = p_j m_j$, the numerator is $p_j(m_j-\mu)$, and substituting gives the second form of κ. Finally, with $\mathbb{1}_j = \mathbb{1}\{y\in C_j\}$,

$(y-\mu)^2 = \sum_j \mathbb{1}_j\{(y-m_j)+(m_j-\mu)\}^2$ and $\mathsf{E}[\mathbb{1}_j(y-m_j)] = 0$, so

$$\mu = \operatorname{var} y = \sum_j \mathsf{E}[\mathbb{1}_j(y-m_j)^2] + \sum_j p_j(m_j-\mu)^2.$$

The first sum is positive: finitely many cells cover the infinitely many points of positive probability, so some cell contains two such points and $y-m_j$ is not almost surely zero on it. Hence $\kappa < 1$. □

Theorem 8.7 (Pearson's statistic with μ estimated by $\bar{y}$)**.** Let $y_1, \dots, y_n$ be independent Pois(μ), let every $p_j(\mu) > 0$, and take $\hat{p}_j = p_j(\bar{y})$. Then, with $\kappa = \kappa(\mu)$ of Lemma 8.6 and $Z_1, \dots, Z_{c-1}$ independent standard normal,

$$X^2 \xrightarrow{d} (1-\kappa)Z_1^2 + \sum_{k=2}^{c-1} Z_k^2.$$

For every x the limit V satisfies $\mathsf{P}(\chi^2_{c-2} > x) \leqslant \mathsf{P}(V > x) \leqslant \mathsf{P}(\chi^2_{c-1} > x)$, χ^2_0 being the point mass at 0.

Proof. Write $p_j = p_j(\mu)$, $a_j = p_j'(\mu)$ and $T_n = n^{-1/2}\sum_i (y_i - \mu)$.

Step 1: linearising the fitted probabilities. $\mathsf{E}\, T_n^2 = \mu$, so T_n is bounded in probability by Chebyshev's inequality and $\bar{y} \xrightarrow{p} \mu$. On $\{\bar{y} > 0\}$, of probability $1 - e^{-n\mu}$, the mean value theorem gives m_{nj} between $\bar{y}$ and μ with

$$\sqrt{n}\{p_j(\bar{y}) - p_j\} = p_j'(m_{nj})T_n = a_j T_n + \{p_j'(m_{nj}) - a_j\}T_n,$$

and the last term tends to 0 in probability: $m_{nj} \xrightarrow{p} \mu$, p_j' is continuous (Lemma 8.6), $T_n \xrightarrow{d} N(0, \mu)$ by the Lindeberg–Lévy corollary, and the continuous mapping theorem and Slutsky's theorem of the prerequisites apply.

Step 2: a sum of independent vectors. Let $W_{nj} = (N_j - np_j(\bar{y}))/\sqrt{np_j}$. By Step 1, $W_{nj} = n^{-1/2}\sum_i \xi_{ij} + o_p(1)$ with

$$\xi_{ij} = \frac{\mathbb{1}\{y_i \in C_j\} - p_j - a_j(y_i - \mu)}{\sqrt{p_j}}.$$

The vectors $\boldsymbol{\xi}_i = (\xi_{i1}, \ldots, \xi_{ic})'$ are independent and identically distributed with mean $\mathbf{0}$. As

$\mathsf{E}[(\mathbb{1}\{y \in C_j\} - p_j)(y - \mu)] = \mu a_j$ by Lemma 8.6 and $\operatorname{var} y = \mu$,

$$\operatorname{cov}(\xi_{ij}, \xi_{il}) = \frac{p_j\delta_{jl} - p_j p_l - 2\mu a_j a_l + \mu a_j a_l}{\sqrt{p_j p_l}}, \qquad \text{so} \qquad \operatorname{var} \boldsymbol{\xi}_i = \mathbf{I}_c - \mathbf{u}\mathbf{u}' - \mathbf{v}\mathbf{v}',$$

with $u_j = \sqrt{p_j}$ and $v_j = \sqrt{\mu}\, a_j/\sqrt{p_j}$. Here $|\mathbf{u}|^2 = \sum_j p_j = 1$, $\mathbf{u}'\mathbf{v} = \sqrt{\mu}\sum_j a_j = 0$ and $|\mathbf{v}|^2 = \kappa$.

Step 3: an orthonormal basis. Let $\mathbf{q}_1 = \mathbf{u}$, let $\mathbf{q}_2 = \mathbf{v}/\sqrt{\kappa}$ if $\kappa > 0$ and any unit vector orthogonal to $\mathbf{u}$ otherwise, and complete to an orthonormal basis $\mathbf{q}_1, \ldots, \mathbf{q}_c$ of $\mathbb{R}^c$. Then $\mathbf{q}_1'\boldsymbol{\xi}_i = \sum_j \{\mathbb{1}\{y_i \in C_j\} - p_j - a_j(y_i - \mu)\} = 1 - 1 - 0 = 0$ identically, and $\boldsymbol{\zeta}_i = (\mathbf{q}_2'\boldsymbol{\xi}_i, \ldots, \mathbf{q}_c'\boldsymbol{\xi}_i)'$ has covariance $\operatorname{diag}(1 - \kappa, 1, \ldots, 1)$, positive definite because $\kappa < 1$. Each coordinate of $\boldsymbol{\xi}_i$ is at most $(1 + |a_j|\,|y_i - \mu|)/\sqrt{p_j}$ in absolute value, so $\mathsf{E}\,|\boldsymbol{\zeta}_i|^3 < \infty$ by Proposition 8.2(ii), and the Lyapunov corollary gives $n^{-1/2}\sum_i \boldsymbol{\zeta}_i \xrightarrow{d} N(\mathbf{0}, \operatorname{diag}(1 - \kappa, 1, \ldots, 1))$.

Step 4: the limit of $|\mathbf{W}_n|^2$. Since $\mathbf{q}_1, \ldots, \mathbf{q}_c$ is an orthonormal basis of $\mathbb{R}^c$, $|\mathbf{W}_n|^2 = \sum_k (\mathbf{q}_k'\mathbf{W}_n)^2$. By Steps 2 and 3, $\mathbf{q}_1'\mathbf{W}_n = o_p(1)$ and $(\mathbf{q}_k'\mathbf{W}_n)_{k \geqslant 2} = n^{-1/2}\sum_i \boldsymbol{\zeta}_i + o_p(1)$, so Slutsky's theorem and the continuous mapping theorem give $|\mathbf{W}_n|^2 \xrightarrow{d} (1 - \kappa)Z_1^2 + \sum_{k=2}^{c-1} Z_k^2$; the limiting coordinates are uncorrelated jointly normal, hence independent by the prerequisites' theorem **uncorrelated blocks are independent**.

Step 5: the denominators. $X^2 = \sum_j W_{nj}^2\, p_j/p_j(\bar{y})$, and $p_j/p_j(\bar{y}) \xrightarrow{p} 1$ by continuity, so $X^2 - |\mathbf{W}_n|^2 \xrightarrow{p} 0$ and Slutsky's theorem finishes the proof. The bounds hold because the limit is $\sum_{k\geqslant 2} Z_k^2$ plus the non-negative $(1-\kappa)Z_1^2$, and at most $\sum_{k\geqslant 1} Z_k^2$. □

Example 8.2 (Singapore, continued; FREES §12.1.1, exercise 12.2)**.** FREES's last cell is $\{4\}$, and its five cells miss $\{5, 6, \ldots\}$; the cells of Theorem 8.7 are $\{0\}, \{1\}, \{2\}, \{3\}, \{4, 5, \ldots\}$:

```
o  <- c(table(factor(pmin(y, 4), levels = 0:4)))
eF <- n * dpois(0:4, ybar)                                  # FREES's cells
eT <- n * c(dpois(0:3, ybar), ppois(3, ybar, lower.tail = FALSE))
X2 <- function(o, e) sum((o - e)^2 / e)
round(c(X2_frees = X2(o, eF), X2_tail_cell = X2(o, eT),
        cell3_term = (o[[4]] - eT[4])^2 / eT[4]), 3)
```

```
##     X2_frees X2_tail_cell   cell3_term
##        41.98        41.98        32.69
```

```
a <- c(dpois(0:3, ybar) * (0:3 / ybar - 1), dpois(3, ybar))  # p_j'(mu)
kap <- ybar * sum(a^2 / (eT / n))
```

```r
signif(c(one_minus_kappa = 1 - kap, p_chisq2 = pchisq(X2(o, eT), 2, lower.tail =
  ↪ FALSE),
         p_chisq3 = pchisq(X2(o, eT), 3, lower.tail = FALSE)), 4)
```

```
## one_minus_kappa        p_chisq2        p_chisq3
##       1.916e-07       7.642e-10       4.043e-09
```

FREES's 41.98 is reproduced, and the tail cell changes it in the fourth decimal place. The cell $\{3\}$, with fitted count 0.40, contributes 32.7 of it. Here $1-\kappa = 1.9\times10^{-7}$, so the limit of Theorem 8.7 is χ^2_3 up to a summand $1.9\times10^{-7}Z_1^2$, and FREES's df $= 5-1-1 = 3$ gives a p-value of order 10^{-9}. That limit fixes the cells and lets $n\to\infty$; at $n = 7483$ the fitted count of cell $\{3\}$ is 0.40. The null law at the actual n, by simulation:

```r
set.seed(812)
Xsim <- replicate(20000, {
  ys <- rpois(n, ybar); m <- mean(ys)
  os <- c(table(factor(pmin(ys, 4), levels = 0:4)))
  X2(os, n * c(dpois(0:3, m), ppois(3, m, lower.tail = FALSE))) })
c(p_simulated = mean(Xsim >= X2(o, eT)),
```

```
  quantile95_simulated = unname(quantile(Xsim, 0.95)), quantile95_chisq3 =
  ↪ qchisq(0.95, 3))
```

```
##          p_simulated quantile95_simulated
##              0.00695              7.14078
##    quantile95_chisq3
##              7.81473
```

The simulated p-value is 0.007, six orders of magnitude above the χ^2_3 value: four counts of 3 in a cell expecting 0.40 are rare, but not as rare as the limit says. The Poisson is still rejected.

1.3 Linear regression on counts

Example 8.3 (Bike sharing; ISLR §4.6.1, Table 4.10)**.** ISLR's `Bikeshare`: $n = 8645$ hours of a Washington, DC bike-sharing programme, with `bikers` the hourly count of users, regressed by least squares on `mnth` and `hr` (factors coded to sum to zero, as in ISLR), `workingday`, `temp` and `weathersit`:

```
B <- Bikeshare
contrasts(B$hr) <- contr.sum(24); contrasts(B$mnth) <- contr.sum(12)
fB <- bikers ~ mnth + hr + workingday + temp + weathersit
```

```r
mlm <- lm(fB, data = B)
round(coef(summary(mlm))[c(1, 36:40), ], 2)
```

```
##                              Estimate Std. Error t value
## (Intercept)                     73.60       5.13   14.34
## workingday                       1.27       1.78    0.71
## temp                           157.21      10.26   15.32
## weathersitcloudy/misty         -12.89       1.96   -6.56
## weathersitlight rain/snow      -66.49       2.97  -22.42
## weathersitheavy rain/snow     -109.74      76.67   -1.43
##                              Pr(>|t|)
## (Intercept)                      0.00
## workingday                       0.48
## temp                             0.00
## weathersitcloudy/misty           0.00
## weathersitlight rain/snow        0.00
## weathersitheavy rain/snow        0.15
```

```r
c(min_bikers = min(B$bikers),
  negative_fitted = mean(fitted(mlm) < 0))
```

```
##      min_bikers negative_fitted
##         1.00000         0.09636
```

```
hr <- as.numeric(as.character(B$hr))
night <- B$bikers[hr %in% 1:3 & B$mnth %in% c("Dec", "Jan", "Feb") &
                    B$weathersit == "light rain/snow"]
morning <- B$bikers[hr %in% 7:9 & B$mnth %in% c("April", "May", "June") &
                      B$weathersit == "clear"]
round(c(mean_night = mean(night), sd_night = sd(night),
        mean_morning = mean(morning), sd_morning = sd(morning)), 2)
```

```
##   mean_night     sd_night mean_morning   sd_morning
##         5.05         3.73       243.59       131.70
```

ISLR's Table 4.10 is reproduced, two entries differing in the last printed digit (-22.42 and -109.74 here, -22.43 and -109.75 in ISLR). 9.6% of the fitted values are negative. ISLR's two subsets, "between 1 AM and 4 AM" and "between 7 AM and 10 AM", are the hours 1–3 and 7–9: means 5.05 and 243.59, standard deviations 3.73 and 131.70.

Proposition 8.8 (Three drawbacks of a linear mean for counts; ISLR §4.6.1)**.** Let $y_1, \dots, y_n$ be counts with $\mathsf{E}\, y_i = \mathbf{x}_i'\boldsymbol{\beta}$.

(i) If $\mathbf{x}'\boldsymbol{\beta} \geqslant 0$ for every $\mathbf{x} = (1, x_1, \dots, x_k)'$ with $x_1, \dots, x_k \in \mathbb{R}$, then $\beta_1 = \cdots = \beta_k = 0$.

(ii) If the y_i are Poisson, $\operatorname{var} y_i = \mathbf{x}_i'\boldsymbol{\beta}$, and A3 holds only if all the means are equal.

(iii) $y_i - \mathbf{x}_i'\boldsymbol{\beta}$ takes only the values $j - \mathbf{x}_i'\boldsymbol{\beta}$, $j = 0, 1, \dots$, so A5 fails.

Proof. (i) If $\beta_j \neq 0$, the vector with $x_j = t$ and the other slopes' coordinates 0 gives $\beta_0 + t\beta_j$, negative for t of the sign opposite to β_j and large. (ii) Proposition 8.2(ii). (iii) A normal variable gives positive probability to every interval, and $y_i - \mathbf{x}_i'\boldsymbol{\beta}$ gives probability 0 to the open interval $(-\mathbf{x}_i'\boldsymbol{\beta}, 1 - \mathbf{x}_i'\boldsymbol{\beta})$ between two of its consecutive values. □

Remark (ISLR §4.6.1)**.** Regressing $\ln y_i$ instead removes negative predictions and much of the heteroscedasticity, but is undefined at $y_i = 0$ and models $\mathsf{E} \ln y$, not $\ln \mathsf{E}\, y$. `Bikeshare` has no zero count, the smallest being 1.

2 Poisson Regression

FREES §12.1.2–12.1.4; ISLR §4.6.2 (PA reading list, not SRM). SRM learning outcomes 2(b), 2(c), 2(e), 2(f).

2.1 The model, exposure and the offset

Definition 8.9 (Poisson regression; FREES (12.4), ISLR (4.36))**.** Given fixed $\mathbf{x}_i = (1, x_{i1}, \ldots, x_{ik})'$, the rows of $\mathbf{X}$, and known **exposures** $E_i > 0$, the **Poisson regression model** takes $y_1, \ldots, y_n$ independent with

$$y_i \sim \text{Pois}(\mu_i), \qquad \ln \mu_i = \ln E_i + \mathbf{x}_i'\boldsymbol{\beta}. \tag{3}$$

The function ln connecting μ_i to $\mathbf{x}_i'\boldsymbol{\beta}$ is the **logarithmic link**; the term $\ln E_i$, a covariate whose coefficient is fixed at 1, is an **offset**. Without exposures, $E_i = 1$.

Proposition 8.10 (Interpretation; FREES §12.1.2, ISLR §4.6.2)**.** Under (3):

(i) $\mu_i = E_i e^{\mathbf{x}_i'\boldsymbol{\beta}} > 0$ for every $\boldsymbol{\beta}$;

(ii) raising x_{ij} by one unit, the other coordinates and E_i fixed, multiplies μ_i by e^{β_j}, and $\beta_j =$

$\partial \ln \mu_i / \partial x_{ij}$ is the proportional change in the mean per unit change of x_{ij};

(iii) the mean is proportional to the exposure;

(iv) with indicator covariates for the levels of two factors, $\mu_i = E_i \cdot B \cdot A_{a(i)} \cdot T_{u(i)}$ with $B = e^{\beta_0}$ and one factor per level: a **multiplicative** structure.

Proof. (i) Exponentiate. (ii) The ratio of the two means is $e^{(x_{ij}+1)\beta_j - x_{ij}\beta_j} = e^{\beta_j}$; differentiate $\ln \mu_i$. (iii) μ_i / E_i does not involve E_i. (iv) With $\mathbf{x}_i'\boldsymbol{\beta} = \beta_0 + \alpha_{a(i)} + \tau_{u(i)}$, exponentiate term by term. □

Remark (Exposure; FREES §12.1.2)**.** By Proposition 8.2(iii) a policy observed for m periods, with independent Pois(μ) counts in each, has a Pois($m\mu$) total: proportionality to exposure is a theorem for integer exposures and an assumption for fractional ones, such as the fraction of the year a Singapore policy was in force. Entering $\ln E_i$ as an ordinary covariate with coefficient γ fits $\mu_i = E_i^{\gamma} e^{\mathbf{x}_i'\boldsymbol{\beta}}$, which drops (iii) unless $\gamma = 1$. Weber's California study (FREES §12.1.2) used the linear mean $\mathsf{E}\, y_i = \mathbf{x}_i'\boldsymbol{\beta}$, which Proposition 8.8(i) restricts.

2.2 Estimation

Theorem 8.11 (Score, information, concavity; FREES §12.1.3)**.** Under (3) the log-likelihood, score and information are

$$L(\boldsymbol{\beta}) = \sum_{i=1}^{n} \left\{ -E_i e^{\mathbf{x}_i'\boldsymbol{\beta}} + y_i(\ln E_i + \mathbf{x}_i'\boldsymbol{\beta}) - \ln y_i! \right\}, \tag{4}$$

$$\nabla L(\boldsymbol{\beta}) = \sum_{i=1}^{n} (y_i - \mu_i)\mathbf{x}_i = \mathbf{X}'(\mathbf{Y} - \boldsymbol{\mu}), \qquad \mathcal{I}(\boldsymbol{\beta}) = \sum_{i=1}^{n} \mu_i \mathbf{x}_i \mathbf{x}_i' = \mathbf{X}'\mathbf{W}\mathbf{X}, \tag{5}$$

with $\mathbf{W} = \operatorname{diag}(\mu_i)$, and $-\nabla^2 L(\boldsymbol{\beta}) = \mathcal{I}(\boldsymbol{\beta})$ whatever the responses. L is concave, and strictly concave when $\operatorname{rank}(\mathbf{X}) = k + 1$.

Proof. Sum the logarithm of (1) at $\mu_i = E_i e^{\mathbf{x}_i'\boldsymbol{\beta}}$. As $\partial\mu_i/\partial\boldsymbol{\beta} = \mu_i\mathbf{x}_i$, $\nabla L = \sum_i (y_i - \mu_i)\mathbf{x}_i$ and $\nabla^2 L = -\sum_i \mu_i \mathbf{x}_i\mathbf{x}_i'$, free of the y_i. The score has mean $\mathbf{0}$ and independent summands of variance $\mu_i\mathbf{x}_i\mathbf{x}_i'$, so its variance is $\sum_i \mu_i\mathbf{x}_i\mathbf{x}_i'$. For $\mathbf{c} \neq \mathbf{0}$, $\mathbf{c}'\mathcal{I}\mathbf{c} = \sum_i \mu_i(\mathbf{x}_i'\mathbf{c})^2$ is non-negative, and positive unless $\mathbf{X}\mathbf{c} = \mathbf{0}$; the prerequisites' lemma **convexity from the derivative**, applied along lines,

gives the concavity and the strict concavity. □

Corollary 8.12 (The score equations; FREES §12.1.3)**.** A maximiser $\mathbf{b}$, with $\hat{\mu}_i = E_i e^{\mathbf{x}_i'\mathbf{b}}$, satisfies $\sum_i x_{ij}\hat{\mu}_i = \sum_i x_{ij} y_i$ for $j = 0, \ldots, k$. With the intercept, $\sum_i (y_i - \hat{\mu}_i) = 0$, and the fitted and observed counts agree in total and within every level of a factor entered by indicators.

Proof. At an interior maximiser of the differentiable L the gradient $\mathbf{X}'(\mathbf{Y} - \hat{\boldsymbol{\mu}})$ vanishes; read it coordinate by coordinate, with $x_{i0} = 1$ and x_{ij} an indicator. □

Theorem 8.13 (Existence)**.** Let $\operatorname{rank}(\mathbf{X}) = k + 1$.

(i) If some $\mathbf{c} \neq \mathbf{0}$ has $\mathbf{x}_i'\mathbf{c} \leqslant 0$ for every i and $\mathbf{x}_i'\mathbf{c} = 0$ whenever $y_i > 0$, no maximum likelihood estimate exists.

(ii) Otherwise L has exactly one maximiser.

Proof. Write $-L(\boldsymbol{\beta}) = \sum_i h_i(\mathbf{x}_i'\boldsymbol{\beta}) + \text{const}$ with $h_i(\eta) = E_i e^{\eta} - y_i \eta$, convex. If $y_i > 0$, h_i decreases on $(-\infty, s_i]$ and increases on $[s_i, \infty)$, where $s_i = \ln(y_i/E_i)$; if $y_i = 0$, h_i increases on $\mathbb{R}$. (i) For any $\boldsymbol{\beta}$, $L(\boldsymbol{\beta} + t\mathbf{c}) = \sum_i \{ y_i(\ln E_i + \mathbf{x}_i'\boldsymbol{\beta}) - E_i e^{\mathbf{x}_i'\boldsymbol{\beta}} e^{t\mathbf{x}_i'\mathbf{c}} - \ln y_i! \}$, as $y_i \mathbf{x}_i'\mathbf{c} = 0$. Some $\mathbf{x}_i'\mathbf{c} < 0$,

since otherwise $\mathbf{X}\mathbf{c} = \mathbf{0}$; so $L(\boldsymbol{\beta} + t\mathbf{c})$ is strictly increasing in t and no $\boldsymbol{\beta}$ maximises L.
(ii) For a unit vector $\mathbf{c}$ put $m(\mathbf{c}) = \max\{\max_i \mathbf{x}_i'\mathbf{c},\ \max_{i:y_i>0}(-\mathbf{x}_i'\mathbf{c})\}$. If $m(\mathbf{c}) \leqslant 0$, every $\mathbf{x}_i'\mathbf{c} \leqslant 0$ and $\mathbf{x}_i'\mathbf{c} = 0$ whenever $y_i > 0$, excluded; so $m > 0$ on the unit sphere. m is continuous, the sphere is compact by the prerequisites' theorem **Heine–Borel**, and m attains a minimum $m_0 > 0$ by the theorem **uniform continuity on compacts**. Let $h_i^* = \inf h_i$, which is 0 if $y_i = 0$ and $h_i(s_i)$ if $y_i > 0$, and let $t > \max_{i:y_i>0} |s_i|/m_0$. If $m(\mathbf{c}) = \mathbf{x}_j'\mathbf{c}$, then $t\mathbf{x}_j'\mathbf{c} \geqslant tm_0 > s_j$ and $h_j(t\mathbf{x}_j'\mathbf{c}) \geqslant h_j(tm_0)$; if $m(\mathbf{c}) = -\mathbf{x}_j'\mathbf{c}$ with $y_j > 0$, then $t\mathbf{x}_j'\mathbf{c} \leqslant -tm_0 < s_j$ and $h_j(t\mathbf{x}_j'\mathbf{c}) \geqslant h_j(-tm_0)$. Bounding every other h_i by h_i^*,

$$-L(t\mathbf{c}) - \text{const} \geqslant \sum_i h_i^* + \min_j \{g_j(t) - h_j^*\},$$

where $g_j(t) = \min\{h_j(tm_0), h_j(-tm_0)\}$ if $y_j > 0$ and $g_j(t) = h_j(tm_0)$ if $y_j = 0$. Each $g_j(t) \to \infty$, since $h_j(tm_0) \geqslant E_j e^{tm_0} - y_j tm_0$ and $h_j(-tm_0) \geqslant y_j tm_0$, and the minimum is over finitely many j, so $-L(t\mathbf{c}) \to \infty$ uniformly in $\mathbf{c}$. Hence $-L$ is continuous and coercive and has a minimiser by the prerequisites' lemma **coercive minimisation**, unique by strict convexity (Theorem 8.11). □

Corollary 8.14 (An empty level)**.** If a factor is entered by indicators against a reference level and some non-reference level, observed at least once, has $y_i = 0$ for every observation in it, no maximum likelihood estimate exists.

Proof. Take $\mathbf{c} = -\mathbf{e}_l$ for that level's indicator l: $\mathbf{x}_i'\mathbf{c} = -1$ on the level and 0 elsewhere, and Theorem 8.13(i) applies. □

2.3 Large-sample inference

Theorem 8.15 (Sampling distribution; FREES §12.1.3)**.** Let (3) hold with true parameter $\boldsymbol{\beta}_0$, and suppose

(D1) $|\mathbf{x}_i| \leqslant M$ and $c_E \leqslant E_i \leqslant C_E$ for constants $M, c_E, C_E > 0$ free of i;

(D2) $\mathbf{F}_n = n^{-1}\sum_{i\leqslant n} \mu_i \mathbf{x}_i\mathbf{x}_i' \to \mathbf{F}$ positive definite, the μ_i evaluated at $\boldsymbol{\beta}_0$.

Then with probability tending to one the maximum likelihood estimate $\mathbf{b}$ exists and is unique, $\mathbf{b} \xrightarrow{p} \boldsymbol{\beta}_0$,

$$\sqrt{n}(\mathbf{b} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{F}^{-1}), \qquad n^{-1}\mathcal{I}(\mathbf{b}) \xrightarrow{p} \mathbf{F},$$

and the Wald and likelihood ratio statistics for q linear restrictions $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ that hold at $\boldsymbol{\beta}_0$ are

χ^2_q in the limit.

Proof. The hypotheses of the prerequisites' theorem **concave maximum likelihood** are checked with $\Theta = \mathbb{R}^{k+1}$ and ℓ_i the ith summand of (4).

Concavity and smoothness. Theorem 8.11.

The score condition. $\nabla \ell_i(\boldsymbol{\beta}_0) = (y_i - \mu_i)\mathbf{x}_i$ has mean $\mathbf{0}$ and variance $\mu_i \mathbf{x}_i \mathbf{x}_i'$, whose average is $\mathbf{F}_n \to \mathbf{F}$. Under (D1) every $\mu_i = E_i e^{\mathbf{x}_i' \boldsymbol{\beta}_0}$ lies in $[m_-, m_+]$ with $m_- = c_E e^{-M|\boldsymbol{\beta}_0|} > 0$ and $m_+ = C_E e^{M|\boldsymbol{\beta}_0|}$, so by Proposition 8.2(ii)

$$\mathsf{E}\,|(y_i - \mu_i)\mathbf{x}_i|^4 \leqslant M^4(m_+ + 3m_+^2),$$

and the Lyapunov corollary with $\delta = 2$ gives $n^{-1/2}\nabla L(\boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{F})$. The summands are unbounded, so the bounded-summand corollary does not apply.

The curvature condition. $\mathbf{H}_n(\boldsymbol{\beta}) = \sum_i E_i e^{\mathbf{x}_i' \boldsymbol{\beta}} \mathbf{x}_i \mathbf{x}_i'$ is non-random and $n^{-1}\mathbf{H}_n(\boldsymbol{\beta}_0) = \mathbf{F}_n \to \mathbf{F}$, so $\mathbf{G} = \mathbf{F}$, the information equality. For $|\boldsymbol{\beta} - \boldsymbol{\beta}_0| \leqslant 1$ the exponents lie in $[-M(|\boldsymbol{\beta}_0|+1), M(|\boldsymbol{\beta}_0|+1)]$, where $e^\eta \leqslant K = e^{M(|\boldsymbol{\beta}_0|+1)}$, so the mean value theorem gives $|e^{\mathbf{x}_i'\boldsymbol{\beta}} - e^{\mathbf{x}_i'\boldsymbol{\beta}_0}| \leqslant KM|\boldsymbol{\beta} - \boldsymbol{\beta}_0|$ and $\|\mathbf{H}_n(\boldsymbol{\beta}) - \mathbf{H}_n(\boldsymbol{\beta}_0)\| \leqslant nC_E K M^3 |\boldsymbol{\beta} - \boldsymbol{\beta}_0|$, a constant Lipschitz bound.

The conclusions are parts (i)–(v) of that theorem with $\mathbf{F} = \mathbf{G}$, unique because L is strictly concave once $\operatorname{rank} \mathbf{X} = k + 1$, which (D2) forces for large n (two maximisers would give a strictly larger value at their midpoint); and its corollary **Wald** and theorem **likelihood ratio**, whose restricted problem satisfies (D1)–(D2) with the corresponding sub-block of $\mathbf{F}$. For restrictions $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ with $\mathbf{R}$ of full row rank q, the likelihood ratio theorem is applied in the coordinates $\boldsymbol{\gamma} = \mathbf{A}^{-1}\boldsymbol{\beta} = (\mathbf{S}\boldsymbol{\beta}, \mathbf{R}\boldsymbol{\beta})$, the rows of $\mathbf{S}$ completing those of $\mathbf{R}$ to a basis: the covariates $\mathbf{A}'\mathbf{x}_i$ keep (D1)–(D2) with information $\mathbf{A}'\mathbf{F}\mathbf{A}$, the null is the coordinate restriction $\boldsymbol{\gamma}_2 = \mathbf{r}$, and the likelihood ratio is unchanged. □

Remark (FREES §12.1.3–12.1.4)**.** The standard error of b_j is the square root of the jth diagonal entry of $\mathcal{I}(\mathbf{b})^{-1} = (\sum_i \hat{\mu}_i \mathbf{x}_i \mathbf{x}_i')^{-1}$, and $b_j / \operatorname{se}(b_j)$ is the t-ratio (R's "z value"). FREES's AIC $= -2L(\mathbf{b}) + 2(k+1)$ and its likelihood ratio statistic LRT $= 2\{L(\mathbf{b}) - L(\mathbf{b}_{\text{Reduced}})\}$ are those of Unit 7.

Remark (Medical malpractice; FREES §12.1.3, Table 12.2)**.** Fournier and McInnes fitted a Poisson regression to the malpractice claims of $n = 9059$ Florida physicians, 1985–1989; FREES prints their coefficients, not their data. The coefficient of the female indicator is -0.432 with standard error 0.082, so a male physician has $e^{0.432} = 1.540$ times the expected number of claims of a female

one with the same other characteristics (Proposition 8.10), and by Theorem 8.15 with the standard error just defined (the prerequisites' corollary **Wald** with $q = 1$) an approximate 95% interval for that factor is $\exp(0.432 \pm 1.96 \times 0.082) = (1.31, 1.81)$. Speciality matters most: surgery's 1.410 multiplies the expected count by $e^{1.410} = 4.10$ relative to the reference speciality.

Example 8.4 (Bike sharing, Poisson; ISLR §4.6.2, Table 4.11)**.**

```
mpo <- glm(fB, data = B, family = poisson)
round(coef(summary(mpo))[c(1, 36:40), ], 2)
```

```
##                            Estimate Std. Error z value
## (Intercept)                    4.12       0.01  683.96
## workingday                     0.01       0.00    7.50
## temp                           0.79       0.01   68.43
## weathersitcloudy/misty        -0.08       0.00  -34.53
## weathersitlight rain/snow     -0.58       0.00 -141.90
## weathersitheavy rain/snow     -0.93       0.17   -5.55
##                            Pr(>|z|)
## (Intercept)                       0
## workingday                        0
```

```
## temp                          0
## weathersitcloudy/misty        0
## weathersitlight rain/snow     0
## weathersitheavy rain/snow     0
```

```
cw <- coef(mpo)[c("weathersitcloudy/misty", "weathersitlight rain/snow")]
round(c(cloudy_vs_clear = exp(cw[[1]]), rain_vs_cloudy = exp(cw[[2]] - cw[[1]]),
        from_rounded = exp(c(-0.08, -0.50))), 4)
```

```
## cloudy_vs_clear  rain_vs_cloudy   from_rounded1
##          0.9275          0.6062          0.9231
##   from_rounded2
##          0.6065
```

ISLR's Table 4.11 is reproduced, one z value differing in the last digit (-141.90 against -141.91). By Proposition 8.10(ii) cloudy skies multiply the mean by $e^{-0.07523} = 0.9275$ and rain multiplies it by a further 0.6062; ISLR's 0.923 and 0.607 exponentiate the rounded -0.08 and -0.50. `workingday` has $z = 7.50$ here and $t = 0.71$ in Example 8.3.

2.4 Pearson residuals and the Pearson statistic

Definition 8.16 (Pearson residuals; FREES §12.1.4)**.** The **Pearson residuals** and the observation-level **Pearson statistic** of a fitted Poisson regression with $k+1$ coefficients are

$$r_i = \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}, \qquad X^2 = \sum_{i=1}^{n} r_i^2, \qquad \hat{\phi} = \frac{X^2}{n-(k+1)}.$$

Proposition 8.17 (What the Pearson statistic estimates; FREES §12.1.4)**.** Let $y_1, \dots, y_n$ be independent with $\mathsf{E}\, y_i = \mu_i = E_i e^{\mathbf{x}_i' \boldsymbol{\beta}_0}$ and $\operatorname{var} y_i = \phi \mu_i$, let (D1) hold, let $\sup_i \mathsf{E}(y_i - \mu_i)^4 < \infty$, and let $\mathbf{b} - \boldsymbol{\beta}_0$ be bounded in probability after multiplication by $\sqrt{n}$. Then $(y_i - \mu_i)/\sqrt{\mu_i}$ has variance exactly ϕ, and $\hat{\phi} \xrightarrow{p} \phi$. Under (3), $\phi = 1$ and $X^2/\{n-(k+1)\} \xrightarrow{p} 1$.

Proof. The variance statement is $\operatorname{var} y_i/\mu_i = \phi$. Write $e_i = y_i - \mu_i$ and $\delta_i = \hat{\mu}_i - \mu_i$, and let $m_- > 0$ and m_+ bound the μ_i below and above (proof of Theorem 8.15). On $|\mathbf{b} - \boldsymbol{\beta}_0| \leqslant 1$ the mean value theorem gives $\Delta = \max_i |\delta_i| \leqslant C_E K M |\mathbf{b} - \boldsymbol{\beta}_0|$, so $\Delta \to 0$ in probability.
The average $\tilde{\phi} = n^{-1} \sum_i e_i^2/\mu_i$ has mean ϕ and variance at most $\sup_i \mathsf{E}\, e_i^4/(n m_-^2)$, so $\tilde{\phi} \xrightarrow{p} \phi$ by

Chebyshev's inequality. On $\{\Delta \leqslant m_-/2\}$, every $\hat{\mu}_i \geqslant m_-/2$ and

$$\frac{(e_i - \delta_i)^2}{\hat{\mu}_i} - \frac{e_i^2}{\mu_i} = -\frac{e_i^2 \delta_i}{\mu_i \hat{\mu}_i} + \frac{\delta_i^2 - 2e_i\delta_i}{\hat{\mu}_i},$$

so the average of these differences is at most $2\Delta m_-^{-2} n^{-1} \sum_i e_i^2 + 2m_-^{-1}(\Delta^2 + 2\Delta n^{-1} \sum_i |e_i|)$ in absolute value. Here $n^{-1}\sum_i e_i^2 \leqslant m_+\tilde{\phi}$ and $n^{-1}\sum_i |e_i| \leqslant (m_+\tilde{\phi})^{1/2}$ by the Cauchy–Schwarz inequality, so the bound is at most a continuous function of $(\Delta, \tilde{\phi})$ that vanishes at $(0, \phi)$; as $(\Delta, \tilde{\phi}) \xrightarrow{p} (0, \phi)$, the continuous mapping theorem sends it to 0 in probability. With $n/\{n - (k+1)\} \to 1$, $\hat{\phi} \xrightarrow{p} \phi$. □

Remark (FREES §12.1.4)**.** Plots of r_i against fitted values or a candidate covariate locate unusual observations and omitted structure, and X^2 is a goodness-of-fit statistic. FREES's "X^2 should be approximately $n - (k+1)$" is Proposition 8.17: the ratio tends to 1.

3 Application: Singapore Automobile Insurance

FREES §12.2. SRM learning outcomes 2(c), 2(d), 2(e).

Example 8.5 (Vehicle and personal characteristics; FREES Tables 12.4–12.5)**.** `VehicleType A` is an automobile; `VAgecat1` groups vehicle age as 0–2, 3–5, 6–10, 11–15, 16+ (codes 2–6); `AgeCat` codes 2–7 are the ages 22–25, 26–35, 36–45, 46–55, 56–65 and 66+; `NCD` is the no-claims discount in percent.

```
S$Auto <- 1 * (S$VehicleType == "A"); table(Auto = S$Auto, count = y)
```

```
##      count
## Auto    0    1    2    3
##    0 3441  184   13    3
##    1 3555  271   15    1
```

```
table(VAgecat1 = S$VAgecat1, count = y)
```

```
##          count
## VAgecat1    0    1    2    3
##         2 4069  313   20    4
```

```
##         3  708   59    4    0
##         4  872   49    3    0
##         5 1133   30    1    0
##         6  214    4    0    0
```

```
SA <- subset(S, Auto == 1)
rbind(n = table(SA$SexInsured), zero = table(SA$SexInsured[SA$Clm_Count == 0]))
```

```
##         F    M
## n     700 3142
## zero  654 2901
```

```
round(100 * prop.table(table(S$VAgecat1, y == 0), 1)[, 2], 1); sum(S$Auto == 0)
```

```
##    2    3    4    5    6
## 92.4 91.8 94.4 97.3 98.2
## [1] 3641
```

```
round(100 * prop.table(table(SA$Clm_Count == 0, SA$AgeCat), 2)[2, ], 1)
```

```
##    2    3    4    5    6    7
## 92.9 91.7 93.2 93.8 89.2 88.2
```

```r
round(100 * prop.table(table(SA$Clm_Count == 0, SA$NCD), 2)[2, ], 1)
```

```
##    0   10   20   30   40   50
## 89.6 91.2 92.8 93.5 94.8 94.4
```

FREES's Tables 12.4 and 12.5 are reproduced: 3842 automobiles, of which 700 have a female insured; the percentage with no claim is lowest (91.8) for vehicles aged 3–5 and highest (98.2) for those aged 16 or more, and rises from 89.6 at `NCD` $= 0$ to 94.8 at 40 (94.4 at 50). Sex and age are recorded for automobiles only; the remaining 3641 vehicles carry `SexInsured U` (three carry `M`) and `AgeCat` 0 (one carries 4), which the model uses only through `Auto` $= 0$.

Example 8.6 (The fitted Poisson model; FREES Table 12.6)**.** FREES's model, from its chapter script (whose reference level `ref = "7"` for `AgeCatF` does not reproduce Table 12.6; `ref = "2"` does): `Female`, `Auto`, the age and discount factors interacted with `Auto`, vehicle age, and the offset $\ln E_i$ (`LNWEIGHT`), E_i being the fraction of the year in force.

```r
S$NCD1F <- relevel(factor(S$NCD), ref = "50")
S$VAgecat1F <- relevel(factor(S$VAgecat1), ref = "6")
S$AgeCatF <- relevel(factor(S$AgeCat), ref = "2")
fS <- Clm_Count ~ Female + Auto + Auto:AgeCatF + Auto:NCD1F + VAgecat1F
```

```r
m3 <- glm(fS, offset = LNWEIGHT, family = poisson, data = S)
lfact <- sum(lfactorial(y))
round(coef(summary(m3))[, c(1, 3)], 3)
```

```
##                Estimate z value
## (Intercept)      -3.306  -6.602
## Female           -0.173  -1.115
## Auto             -0.667  -1.869
## VAgecat1F2        1.674   3.276
## VAgecat1F3        1.504   2.917
## VAgecat1F4        1.081   2.084
## VAgecat1F5        0.362   0.682
## Auto:AgeCatF3     0.165   0.523
## Auto:AgeCatF4     0.124   0.385
## Auto:AgeCatF5    -0.057  -0.161
## Auto:AgeCatF6     0.489   1.251
## Auto:AgeCatF7     0.747   0.961
## Auto:NCD1F0       0.729   4.704
## Auto:NCD1F10      0.528   2.732
## Auto:NCD1F20      0.293   1.326
```

```
## Auto:NCD1F30     0.260   1.152
## Auto:NCD1F40    -0.095  -0.342
```

```
print(c(logLik = as.numeric(logLik(m3)), sum_log_y_factorial = lfact,
        logLik_plus_sum = as.numeric(logLik(m3)) + lfact), digits = 7)
```

```
##              logLik sum_log_y_factorial     logLik_plus_sum
##        -1803.30552            26.57516         -1776.73036
```

Every estimate and t-ratio of FREES's Table 12.6 is reproduced, with two corrections to the table's labels. The reference age level is 22–25, not "66 and over" as its footnote says, and its five age rows are printed in reverse: FREES's "22–25" row (0.747, $t = 0.961$) is the level 66+, and its "56–65" row (0.165) is the level 26–35. The level `AgeCat` 0 occurs only with `Auto` $= 0$, so `Auto:AgeCatF0` is identically zero and the indicator is dropped. FREES's $L(\mathbf{b}) = -1776.730$ omits the constant $-\sum_i \ln y_i! = -26.575$ of (4); R's -1803.306 includes it. The constant cancels in every likelihood ratio.

Example 8.7 (Interpretation and a likelihood ratio test; FREES §12.2)**.**

```
b3 <- coef(m3)
round(c(NCD0_vs_50 = exp(b3[["Auto:NCD1F0"]]),
```

```
        NCD0_vs_30 = exp(b3[["Auto:NCD1F0"]] - b3[["Auto:NCD1F30"]]),
        frees_0_minus_20 = exp(b3[["Auto:NCD1F0"]] - b3[["Auto:NCD1F20"]])), 3)
```

```
##       NCD0_vs_50       NCD0_vs_30 frees_0_minus_20
##            2.074            1.599            1.547
```

```
m3r <- glm(Clm_Count ~ Auto:NCD1F + VAgecat1F, offset = LNWEIGHT,
           family = poisson, data = S)
LRT <- 2 * (as.numeric(logLik(m3)) - as.numeric(logLik(m3r)))
dfr <- df.residual(m3r) - df.residual(m3)
round(c(LRT = LRT, df = dfr, p_value = pchisq(LRT, dfr, lower.tail = FALSE),
        p_frees_df7 = pchisq(LRT, 7, lower.tail = FALSE)), 3)
```

```
##         LRT          df     p_value p_frees_df7
##       5.379       6.000       0.496       0.614
```

An insured with `NCD` $= 0$ has $e^{0.7293} = 2.074$ times the expected claims of one with 50, by Proposition 8.10(ii). Against `NCD` $= 30$ the factor is $e^{0.7293-0.2597} = 1.599$; FREES's 1.55 subtracts 0.293, the coefficient of `NCD` $= 20$. FREES drops **Auto**, **Female** and the five age coefficients and refers LRT $= 5.379$ to χ^2_7. The reduced model that reproduces FREES's $L(\mathbf{b}_{\text{Reduced}}) = -1779.420$ (its script fits no reduced Poisson model) keeps **Auto:NCD1F**, which without the main effect has

six columns, one per discount level, and their sum is `Auto`; so the reduced model is the full one with six restrictions, not seven, and by Theorem 8.15 the reference law is χ^2_6: $p = 0.496$, against 0.614 from χ^2_7 (FREES prints 0.618).

Example 8.8 (Comparing fitted frequency models; FREES Table 12.7)**.** For a fitted model the fitted count of cell j is $\sum_i \mathsf{P}(y_i \in C_j; \hat{\mu}_i)$:

```
cells <- function(P) { e <- colSums(P); c(round(e, 2), X2 = round(X2(o, e), 2)) }
Ppois <- function(mu) sapply(0:4, function(j) dpois(j, mu))
m1 <- glm(Clm_Count ~ 1, family = poisson, data = S)
m2 <- glm(Clm_Count ~ 1, offset = LNWEIGHT, family = poisson, data = S)
rbind(no_exposure = cells(Ppois(fitted(m1))), exposure = cells(Ppois(fitted(m2))),
      poisson = cells(Ppois(fitted(m3))))
```

```
##                                          X2
## no_exposure 6978 487.7 17.04 0.40 0.01 41.98
## exposure    6983 477.7 21.52 0.73 0.02 17.62
## poisson     6987 470.3 24.63 1.09 0.04  8.77
```

The three Poisson columns of FREES's Table 12.7 and their statistics 41.98, 17.62 and 8.77 are reproduced, with FREES's five cells. FREES refers 8.77 to χ^2_4, $p = 0.067$, although its own rule,

cells minus one minus estimated parameters, gives $5 - 1 - 17 < 0$. Neither Theorem 8.5 nor Theorem 8.7 covers seventeen coefficients estimated from the individual counts. The null law, by a parametric bootstrap from the fitted model:

```
set.seed(1207); mu3 <- fitted(m3); Sb <- S
Xb <- replicate(400, {
  Sb$Clm_Count <- rpois(n, mu3); yb <- Sb$Clm_Count
  mb <- glm(fS, offset = LNWEIGHT, family = poisson, data = Sb)
  ob <- c(table(factor(pmin(yb, 4), levels = 0:4)))
  X2(ob, colSums(Ppois(fitted(mb)))) })
round(c(observed = X2(o, colSums(Ppois(mu3))), p_bootstrap = mean(Xb >= 8.7658),
        p_chisq4 = pchisq(8.7658, 4, lower.tail = FALSE)), 3)
```

```
##    observed p_bootstrap    p_chisq4
##       8.766       0.055       0.067
```

The bootstrap p-value is 0.055 against χ^2_4's 0.067.

4 Overdispersion and the Negative Binomial

FREES §12.3; ISLR §4.6.2 (footnote on overdispersion; PA reading list, not SRM). SRM learning outcomes 2(a), 2(d), 2(e).

4.1 Quasi-likelihood and robust standard errors

Definition 8.18 (Over- and underdispersion; FREES §12.3)**.** Counts with $\operatorname{var} y_i > \mathsf{E}\, y_i$ are **overdispersed**, with $\operatorname{var} y_i < \mathsf{E}\, y_i$ **underdispersed**, and with equality **equidispersed**.

Theorem 8.19 (Quasi-Poisson; FREES (12.8)–(12.9))**.** Let $y_1, \ldots, y_n$ be independent with

$$\mathsf{E}\, y_i = \mu_i = E_i e^{\mathbf{x}_i' \boldsymbol{\beta}_0}, \qquad \operatorname{var} y_i = \phi \mu_i, \tag{6}$$

$\phi > 0$, $\sup_i \mathsf{E}(y_i - \mu_i)^4 < \infty$, and (D1), (D2) of Theorem 8.15. Let $\mathbf{b}$ maximise the Poisson log-likelihood (4), whatever the law of the y_i. Then

(i) $\mathbf{b}$ depends on the data only through the score equations $\mathbf{X}'(\mathbf{Y} - \boldsymbol{\mu}) = \mathbf{0}$;

(ii) $\mathbf{b} \xrightarrow{p} \boldsymbol{\beta}_0$ and $\sqrt{n}(\mathbf{b} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \phi \mathbf{F}^{-1})$;

(iii) consistent standard errors are $\sqrt{\hat\phi}$ times the Poisson ones, $\hat\phi$ of Definition 8.16.

Proof. (i) is Corollary 8.12: nothing beyond (4) enters the computation.
(ii) Apply the prerequisites' theorem **concave maximum likelihood** with ℓ_n the Poisson log-likelihood, now a criterion rather than the log-likelihood of the data. Concavity and the curvature condition are those of Theorem 8.15, since $\mathbf{H}_n$ involves no y_i; so $\mathbf{G} = \mathbf{F}$. The score $\sum_i (y_i - \mu_i)\mathbf{x}_i$ has mean $\mathbf{0}$ because the **mean** is correctly specified, and variance $\sum_i \phi\mu_i \mathbf{x}_i\mathbf{x}_i'$, so its limiting average variance is $\phi\mathbf{F}$; the Lyapunov corollary applies with $\delta = 2$ under the assumed fourth-moment bound. Part (iii) of that theorem gives the sandwich $\mathbf{G}^{-1}(\phi\mathbf{F})\mathbf{G}^{-1} = \phi\mathbf{F}^{-1}$.
(iii) $\hat\phi \xrightarrow{p} \phi$ by Proposition 8.17, $\sqrt{n}(\mathbf{b} - \boldsymbol{\beta}_0)$ being bounded in probability by (ii). By part (v) of the theorem, $n^{-1}\mathcal{I}(\mathbf{b}) \xrightarrow{p} \mathbf{F}$, so $\hat\phi\,\mathcal{I}(\mathbf{b})^{-1}$ estimates $\phi\mathbf{F}^{-1}/n$ consistently. □

Proposition 8.20 (Robust standard errors; FREES §12.3)**.** If only $\mathsf{E}\, y_i = \mu_i = E_i e^{\mathbf{x}_i'\boldsymbol{\beta}_0}$ is assumed, with $\operatorname{var} y_i = \tau_i^2$ unrestricted, $\sup_i \mathsf{E}\,|y_i - \mu_i|^4 < \infty$, (D1), (D2), and $n^{-1}\sum_i \tau_i^2 \mathbf{x}_i\mathbf{x}_i' \to \mathbf{F}_\tau$ positive definite, then $\sqrt{n}(\mathbf{b} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{F}^{-1}\mathbf{F}_\tau\mathbf{F}^{-1})$, consistently estimated after division by n

by

$$\widehat{\operatorname{var}}\,\mathbf{b} = \Big(\sum_i \hat{\mu}_i \mathbf{x}_i \mathbf{x}_i'\Big)^{-1} \Big(\sum_i (y_i - \hat{\mu}_i)^2 \mathbf{x}_i \mathbf{x}_i'\Big) \Big(\sum_i \hat{\mu}_i \mathbf{x}_i \mathbf{x}_i'\Big)^{-1}. \tag{7}$$

Proof. The proof of Theorem 8.19(ii) runs unchanged with score variance $\sum_i \tau_i^2 \mathbf{x}_i \mathbf{x}_i'$, giving the stated sandwich. For the middle matrix, $n^{-1}\sum_i e_i^2 \mathbf{x}_i \mathbf{x}_i'$ has mean $n^{-1}\sum_i \tau_i^2 \mathbf{x}_i \mathbf{x}_i'$ and entries of variance at most $M^4 \sup_i \mathsf{E}\, e_i^4/n$, so it converges to $\mathbf{F}_\tau$ by Chebyshev's inequality. Replacing e_i by $e_i - \delta_i$ changes it by at most $M^2(2\Delta A_n + \Delta^2)$ in each entry, $A_n = n^{-1}\sum_i |e_i|$, with $\Delta \xrightarrow{p} 0$ as in Proposition 8.17. As $\mathsf{E}\, A_n \leqslant \sup_i (\mathsf{E}\, e_i^4)^{1/4}$, Markov's inequality gives $\mathsf{P}(\Delta A_n > \varepsilon) \leqslant \mathsf{E}\, A_n/c + \mathsf{P}(\Delta > \varepsilon/c)$ for every $c > 0$, so the change tends to 0 in probability. The outer matrices are $n \cdot n^{-1}\mathcal{I}(\mathbf{b})$, handled by part (v) of the prerequisites' theorem **concave maximum likelihood**, as in the proof of Theorem 8.19(iii). □

Remark (FREES §12.3)**.** FREES prints the quasi-likelihood variance as $(\hat{\phi}\sum_i \hat{\mu}_i \mathbf{x}_i \mathbf{x}_i')^{-1}$, which divides by $\hat{\phi}$; Theorem 8.19(iii) multiplies, $\hat{\phi}(\sum_i \hat{\mu}_i \mathbf{x}_i \mathbf{x}_i')^{-1}$. Quasi-Poisson forces the ratio of robust to model-based standard error to be $\sqrt{\hat{\phi}}$ for every coefficient; (7) does not, and a wide spread of

those ratios is evidence against $\operatorname{var} y_i = \phi\mu_i$ itself. A likelihood ratio statistic computed from a quasi-Poisson fit is twice a difference of criterion values and is divided by $\hat{\phi}$ before comparison with χ^2. Indeed, Step 6 of the proof of the prerequisites' theorem **concave maximum likelihood** gives $2\{\ell_n(\mathbf{b}) - \ell_n(\boldsymbol{\beta}_0)\} = \mathbf{Z}_n'\mathbf{F}^{-1}\mathbf{Z}_n + o_p(1)$ without using that the score variance equals $\mathbf{G}$; here $\mathbf{G} = \mathbf{F}$ and $\mathbf{Z}_n = n^{-1/2}\sum_i (y_i - \mu_i)\mathbf{x}_i \xrightarrow{d} Z \sim N(\mathbf{0}, \phi\mathbf{F})$ by the proof of Theorem 8.19(ii). Fixing q coordinates of $\boldsymbol{\beta}$ at their true values absorbs them into the exposures, which keeps (D1) and (D2) with $\mathbf{F}$ replaced by its leading block $\mathbf{F}_{11}$, so the restricted fit has the same expansion with $\mathbf{Z}_{n,1}$ and $\mathbf{F}_{11}$. The algebra of the proof of the prerequisites' theorem **likelihood ratio**, applied to $Z/\sqrt{\phi} \sim N(\mathbf{0}, \mathbf{F})$, then shows that twice the difference of the two criterion values converges in law to ϕ times a χ^2_q variable, and Slutsky's theorem with $\hat{\phi} \xrightarrow{p} \phi$ (Proposition 8.17, its hypothesis supplied by Theorem 8.19(ii)) gives convergence of the ratio to χ^2_q.

Example 8.9 (Overdispersion in the two data sets; FREES §12.3, ISLR §4.6.2)**.**

```
sandwich_ratio <- function(m) {
  X <- model.matrix(m)[, !is.na(coef(m))]; mu <- fitted(m); e <- m$y - mu
  G <- crossprod(X, mu * X); Fm <- crossprod(X, e^2 * X)
```

```
  sqrt(diag(solve(G, Fm) %*% solve(G))) / sqrt(diag(solve(G))) }
phi_hat <- function(m) sum((m$y - fitted(m))^2 / fitted(m)) / df.residual(m)
q3 <- glm(fS, offset = LNWEIGHT, family = quasipoisson, data = S)
qB <- glm(fB, data = B, family = quasipoisson)
rbind(singapore = c(phi_hat = phi_hat(m3), sqrt_phi = sqrt(phi_hat(m3)),
                    range(sandwich_ratio(m3))),
      bikeshare = c(phi_hat(mpo), sqrt(phi_hat(mpo)), range(sandwich_ratio(mpo))))
```

```
##           phi_hat sqrt_phi
## singapore  0.9968   0.9984 0.9385 1.073
## bikeshare 25.5989   5.0595 0.1564 7.857
```

```
c(coef_diff = max(abs(coef(q3) - coef(m3)), na.rm = TRUE),
  se_ratio_minus_sqrt_phi = max(abs(coef(summary(qB))[, 2] / coef(summary(mpo))[, 2]
   ↪  -
                                    sqrt(summary(qB)$dispersion))))
```

```
##              coef_diff se_ratio_minus_sqrt_phi
##              0.000e+00               8.882e-16
```

On the Singapore fit $\hat{\phi} = 0.997$ and the robust ratios lie between 0.94 and 1.07: given the covariates and the exposure, nothing contradicts $\operatorname{var} y_i = \mu_i$. On **Bikeshare**, $\hat{\phi} = 25.6$, so every Poisson

standard error of Example 8.4 is too small by the factor 5.06 under (6), and the robust ratios run from 0.16 to 7.86, so $\operatorname{var} y_i = \phi\mu_i$ fails as well. The quasi-Poisson coefficients equal the Poisson ones and its standard errors are exactly $\sqrt{\hat\phi}$ times as large, Theorem 8.19(i) and (iii).

4.2 The negative binomial

Definition 8.21 (Negative binomial; FREES (12.10), Table 13.8)**.** For $r > 0$ and $p \in (0,1)$, $y \sim \mathrm{NB}(r,p)$ if

$$\mathsf{P}(y=j) = \frac{\Gamma(j+r)}{j!\,\Gamma(r)}\,p^r(1-p)^j, \qquad j = 0,1,2,\ldots, \tag{8}$$

which is $\binom{j+r-1}{r-1}p^r(1-p)^j$ for integer r. With $\mu = r(1-p)/p$ and $\sigma = 1/r$ the same law is written $\mathrm{NB}(\mu,\sigma)$; then $p = 1/(1+\sigma\mu)$.

Theorem 8.22 (Poisson–gamma mixture; FREES §12.3)**.** Let λ have the gamma density with shape r and rate r/μ, so that $\mathsf{E}\,\lambda = \mu$ and $\operatorname{var}\lambda = \mu^2/r$, and let y given λ be $\mathrm{Pois}(\lambda)$. Then

$y \sim \mathrm{NB}(r, p)$ with $p = r/(r+\mu)$, and

$$\mathsf{E}\, y = \mu = \frac{r(1-p)}{p}, \qquad \operatorname{var} y = \mu + \frac{\mu^2}{r} = \frac{r(1-p)}{p^2}. \tag{9}$$

In particular (8) sums to 1 for every $r > 0$, $p \in (0,1)$.

Proof. Write $\rho = r/\mu$. By Fubini–Tonelli,

$$\mathsf{P}(y = j) = \int_0^\infty \frac{\lambda^j e^{-\lambda}}{j!} \frac{\rho^r}{\Gamma(r)} \lambda^{r-1} e^{-\rho\lambda}\, \mathrm{d}\lambda = \frac{\rho^r}{j!\,\Gamma(r)} \int_0^\infty \lambda^{j+r-1} e^{-(1+\rho)\lambda}\, \mathrm{d}\lambda,$$

and the substitution $u = (1+\rho)\lambda$ turns the integral into $\Gamma(j+r)/(1+\rho)^{j+r}$. Since $\rho/(1+\rho) = r/(r+\mu) = p$ and $1/(1+\rho) = 1-p$, this is (8). Every $p \in (0,1)$ arises from $\mu = r(1-p)/p$, so (8) is a mass function. The tower property and the conditional variance decomposition of the prerequisites give $\mathsf{E}\, y = \mathsf{E}\, \lambda = \mu$ and $\operatorname{var} y = \mathsf{E}\, \lambda + \operatorname{var} \lambda = \mu + \mu^2/r$; substituting $\mu = r(1-p)/p$ gives the (r, p) forms. □

Corollary 8.23 (The Poisson is a limit)**.** For fixed μ and j, $\mathsf{P}(y = j) \to \mu^j e^{-\mu}/j!$ as $r \to \infty$, that is, as $\sigma \to 0$ with $p = r/(r+\mu) \to 1$; and $\operatorname{var} y \downarrow \mu$.

Proof. The variance is immediate from (9). Write (8) with $p = r/(r+\mu)$ as

$$\frac{\Gamma(j+r)}{\Gamma(r)r^j} \cdot \frac{\mu^j}{j!} \cdot \left(1+\frac{\mu}{r}\right)^{-r} \left(1+\frac{\mu}{r}\right)^{-j}.$$

The first factor is $\prod_{l=0}^{j-1}(r+l)/r \to 1$, using $\Gamma(x+1) = x\Gamma(x)$; the third tends to $e^{-\mu}$ and the fourth to 1. □

Remark (FREES §12.3)**.** FREES's limit "$p \to 1$ and $r \to 0$ such that $rp \to \lambda$" is degenerate: $rp \leqslant r \to 0$ forces $\lambda = 0$, and with $r \to 0$ at fixed mean the variance $\mu + \mu^2/r$ diverges. The Poisson is the limit $r \to \infty$ of Corollary 8.23, on the boundary of the parameter space $\sigma \in (0, \infty)$ rather than inside it; the prerequisites' theorem **likelihood ratio** assumes an interior null value, so it does not license a χ^2_1 test of Poisson against negative binomial, as FREES notes.

Definition 8.24 (Negative binomial regression; FREES §12.3)**.** The y_i are independent $\mathrm{NB}(\mu_i, \sigma)$ with $\ln \mu_i = \ln E_i + \mathbf{x}_i'\boldsymbol{\beta}$ and one $\sigma > 0$; by (9), $\operatorname{var} y_i = \mu_i + \sigma\mu_i^2$.

Proposition 8.25 (Concavity in $\boldsymbol{\beta}$ for fixed σ)**.** For fixed σ the negative binomial log-likelihood is concave in $\boldsymbol{\beta}$, strictly when $\operatorname{rank}\mathbf{X} = k+1$, with score $\sum_i (y_i - \mu_i)\mathbf{x}_i/(1+\sigma\mu_i)$.

Proof. With $r = 1/\sigma$, the terms of $\ln \mathsf{P}(y_i = j)$ involving μ_i are $y_i \ln \mu_i - (y_i + r)\ln(r + \mu_i)$. As a function of $\eta = \ln \mu_i$, with $\mathrm{d}\mu_i/\mathrm{d}\eta = \mu_i$, the first derivative is $y_i - (y_i + r)\mu_i/(r+\mu_i) = r(y_i - \mu_i)/(r+\mu_i) = (y_i - \mu_i)/(1+\sigma\mu_i)$ and the second is $-(y_i + r)r\mu_i/(r+\mu_i)^2 < 0$ (the prerequisites' lemma **convexity from the derivative**). Composing with the linear map $\boldsymbol{\beta} \mapsto \ln E_i + \mathbf{x}_i'\boldsymbol{\beta}$ gives the score and preserves concavity, and the Hessian $-\sum_i \{(y_i+r)r\mu_i/(r+\mu_i)^2\}\mathbf{x}_i\mathbf{x}_i'$ is negative definite when the $\mathbf{x}_i$ span $\mathbb{R}^{k+1}$. □

Example 8.10 (Singapore, negative binomial; FREES §12.3, Table 12.7)**.** FREES's script fits the Table 12.6 model by `MASS::glm.nb`, whose `theta` is r:

```
nb3 <- glm.nb(update(fS, . ~ . + offset(LNWEIGHT)), data = S)
print(c(r = nb3$theta, se_r = nb3$SE.theta, logLik = as.numeric(logLik(nb3)),
```

```
        logLik_frees_convention = as.numeric(logLik(nb3)) + lfact,
        AIC_poisson = AIC(m3), AIC_negbin = AIC(nb3)), digits = 7)
```

```
##                      r                     se_r
##               2.331455                1.275526
##                 logLik logLik_frees_convention
##          -1801.069077            -1774.493918
##           AIC_poisson              AIC_negbin
##           3640.611046            3638.138154
```

```
Pnb <- function(mu, r) sapply(0:4, function(j) dnbinom(j, size = r, mu = mu))
rbind(negbin = cells(Pnb(fitted(nb3), nb3$theta)),
      frees_size_2.33 = cells(Pnb(fitted(nb3), 2.33)))
```

```
##                                          X2
## negbin           6997 451.6 31.71 2.42 0.2 1.69
## frees_size_2.33 6997 451.6 31.72 2.42 0.2 1.69
```

$\hat{r} = 2.33$ and, in FREES's convention without $\sum \ln y_i!$, the log-likelihood -1774.494 of FREES §12.3 are reproduced. FREES's negative binomial column of Table 12.7 (6996.04, 453.40, 31.09, 2.28, 0.18; statistic 1.79) is not: the model of its script gives 6997.00, 451.65, 31.71, 2.42, 0.20 and

statistic 1.69, and the script's fixed `size = 2.33` changes the second decimal only. With $\hat{\mu}_i$ near 0.07, $\sigma\hat{\mu}_i^2$ is small beside $\hat{\mu}_i$, yet the extra variance moves fitted mass from 1 to 0, 2 and 3, which is where the Poisson column missed.

5 Other Count Models

FREES §12.4. SRM learning outcomes 2(a), 2(e).

5.1 Zero-inflated and hurdle models

Definition 8.26 (Zero-inflated and hurdle models; FREES (12.11)–(12.12))**.** Let g_i be a count mass function with mean μ_i and $\pi_i \in (0,1)$.

(i) The **zero-inflated** model mixes a point mass at 0 with g_i:

$$\mathsf{P}(y_i = j) = \begin{cases} \pi_i + (1-\pi_i)g_i(0), & j = 0, \\ (1-\pi_i)g_i(j), & j \geqslant 1. \end{cases} \tag{10}$$

(ii) The **hurdle** model fixes the zero probability and rescales the rest, for $g_i(0) < 1$:

$$\mathsf{P}(y_i = j) = \begin{cases} \pi_i, & j = 0, \\ k_i\, g_i(j), & j \geqslant 1, \end{cases} \qquad k_i = \frac{1-\pi_i}{1-g_i(0)}. \tag{11}$$

In both, π_i is usually a logistic regression and μ_i a log-linear one.

Theorem 8.27 (Moments; FREES §12.4.1–12.4.2)**.** Let g_i be the Pois(μ_i) mass function.

(i) Under (10), $\mathsf{E}\, y_i = (1-\pi_i)\mu_i$ and $\operatorname{var} y_i = (1-\pi_i)\mu_i + \pi_i(1-\pi_i)\mu_i^2$, which exceeds the mean.

(ii) Under (11), $\mathsf{E}\, y_i = k_i\mu_i$ and $\operatorname{var} y_i = k_i\mu_i + k_i(1-k_i)\mu_i^2$, which exceeds the mean if $k_i < 1$ and falls below it if $k_i > 1$.

Proof. (i) Let D be a Bernoulli(π_i) indicator independent of $N \sim$ Pois(μ_i); then $(1-D)N$ has law (10). As $(1-D)^2 = 1-D$, $\mathsf{E}\, y_i = (1-\pi_i)\mu_i$ and $\mathsf{E}\, y_i^2 = (1-\pi_i)(\mu_i + \mu_i^2)$, so $\operatorname{var} y_i = (1-\pi_i)\mu_i + (1-\pi_i)\mu_i^2 - (1-\pi_i)^2\mu_i^2$, and the last two terms combine to $\pi_i(1-\pi_i)\mu_i^2 > 0$. (ii) The $j = 0$ terms contribute nothing to $\mathsf{E}\, y_i = k_i \sum_{j\geqslant 1} j g_i(j) = k_i\mu_i$ or to $\mathsf{E}\, y_i^2 = k_i(\mu_i + \mu_i^2)$;

subtract the squared mean. □

Remark (FREES §12.4.1)**.** FREES prints the zero-inflated Poisson variance as $\pi_i\mu_i+\pi_i\mu_i^2(1-\pi_i)$; the first term is $(1-\pi_i)\mu_i$, as in Theorem 8.27(i). FREES's conclusion, that the variance exceeds the mean, holds for the corrected formula.

Theorem 8.28 (The hurdle likelihood separates; FREES §12.4.2)**.** Let $\pi_i = \pi(\mathbf{x}_i;\boldsymbol{\gamma})$ and $g_i = g(\cdot;\mathbf{x}_i,\boldsymbol{\beta})$ with $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$ variation-independent. The log-likelihood of (11) is

$$L(\boldsymbol{\gamma},\boldsymbol{\beta}) = \underbrace{\sum_i \big[\mathbb{1}\{y_i=0\}\ln\pi_i + \mathbb{1}\{y_i>0\}\ln(1-\pi_i)\big]}_{L_1(\boldsymbol{\gamma})} + \underbrace{\sum_{i:y_i>0} \ln\frac{g_i(y_i)}{1-g_i(0)}}_{L_2(\boldsymbol{\beta})},$$

so it is maximised by maximising the binary log-likelihood L_1 and the zero-truncated count log-likelihood L_2 separately.

Proof. An observation with $y_i = 0$ contributes $\ln \pi_i$, and one with $y_i > 0$ contributes $\ln k_i + \ln g_i(y_i) = \ln(1-\pi_i) + \ln\{g_i(y_i)/(1-g_i(0))\}$. No parameter appears in both sums, so $\sup L = \sup L_1 + \sup L_2$, attained at any pair of separate maximisers. □

Corollary 8.29 (An intercept-only hurdle reproduces the zeros)**.** If $\pi_i = e^{\gamma_0}/(1+e^{\gamma_0})$ for all i, with $0 < \#\{y_i = 0\} < n$, then $\hat{\pi} = n^{-1}\#\{i : y_i = 0\}$ and the fitted number of zeros equals the observed number.

Proof. L_1 is $N_0 \ln \pi + (n - N_0)\ln(1-\pi)$ with $N_0 = \#\{y_i = 0\}$, composed with the strictly increasing $\gamma_0 \mapsto \pi$; its derivative in π, $N_0/\pi - (n-N_0)/(1-\pi)$, is strictly decreasing and vanishes at $\pi = N_0/n$, the maximiser, and the fitted number of zeros is $n\hat{\pi} = N_0$. □

5.2 Heterogeneity and latent class models

Proposition 8.30 (Heterogeneity; FREES §12.4.3)**.** Let y_i given α_i be $\mathrm{Pois}(e^{\alpha_i}\mu_i)$, where $\mathsf{E}\, e^{\alpha_i} = 1$ and $\operatorname{var} e^{\alpha_i} = \tau^2 < \infty$. Then $\mathsf{E}\, y_i = \mu_i$ and $\operatorname{var} y_i = \mu_i + \tau^2\mu_i^2$. If e^{α_i} is gamma with shape r and rate r, $y_i \sim \mathrm{NB}(\mu_i, 1/r)$.

Proof. By the tower property and the conditional variance decomposition, $\mathsf{E}\, y_i = \mu_i \,\mathsf{E}\, e^{\alpha_i} = \mu_i$ and $\operatorname{var} y_i = \mathsf{E}(e^{\alpha_i}\mu_i) + \operatorname{var}(e^{\alpha_i}\mu_i) = \mu_i + \tau^2\mu_i^2$. In the gamma case $\lambda = e^{\alpha_i}\mu_i$ is gamma with shape r and rate r/μ_i, and Theorem 8.22 applies. □

Proposition 8.31 (Latent classes; FREES (12.13))**.** Let $\mathsf{P}(y = j) = \sum_{c=1}^{m} \pi_c \,\mathsf{P}_c(y = j)$ with weights $\pi_c > 0$ summing to 1 and component means μ_c and variances v_c. Then $\operatorname{var} y = \sum_c \pi_c v_c + \sum_c \pi_c(\mu_c - \bar{\mu})^2$ with $\bar{\mu} = \sum_c \pi_c \mu_c$. If every component has $v_c \geqslant \mu_c$, as the Poisson and the negative binomial do, then $\operatorname{var} y \geqslant \mathsf{E}\, y$, strictly unless every $v_c = \mu_c$ and all μ_c are equal.

Proof. Let C be the class, with $\mathsf{P}(C = c) = \pi_c$. The conditional variance decomposition gives $\operatorname{var} y = \mathsf{E}\, v_C + \operatorname{var} \mu_C$, which is the displayed sum, and $\mathsf{E}\, y = \bar{\mu}$. Then $\operatorname{var} y - \mathsf{E}\, y = \sum_c \pi_c(v_c - \mu_c) + \sum_c \pi_c(\mu_c - \bar{\mu})^2$, a sum of non-negative terms. □

Remark (FREES §12.4.4)**.** FREES says a latent class model "readily accommodates under- and overdispersion". With Poisson or negative binomial classes, as in its examples, it cannot produce underdispersion (Proposition 8.31); among the models of this section only the hurdle can (Theorem 8.27(ii)).

Example 8.11 (Excess zeros in the Singapore data; FREES §12.4)**.** The Table 12.6 count part with an intercept-only zero part, fitted by **pscl**:

```
suppressMessages(library(pscl))
S$X3 <- model.matrix(m3)[, !is.na(coef(m3))][, -1]
zi <- zeroinfl(Clm_Count ~ X3 + offset(LNWEIGHT) | 1, data = S, dist = "poisson")
hu <- hurdle(Clm_Count ~ X3 + offset(LNWEIGHT) | 1, data = S, dist = "poisson")
round(c(AIC_poisson = AIC(m3), AIC_negbin = AIC(nb3), AIC_zip = AIC(zi),
        AIC_hurdle = AIC(hu), pi_zip = plogis(coef(zi)[["zero_(Intercept)"]])), 3)
```

```
## AIC_poisson  AIC_negbin     AIC_zip  AIC_hurdle      pi_zip
##    3640.611    3638.138    3638.798    3870.800       0.279
```

```
round(c(observed = sum(y == 0), poisson = sum(dpois(0, fitted(m3))),
        negbin = sum(dnbinom(0, size = nb3$theta, mu = fitted(nb3))),
        zip = sum(predict(zi, type = "prob")[, 1]),
        hurdle = sum(predict(hu, type = "prob")[, 1])), 2)
```

```
## observed  poisson   negbin      zip   hurdle
##     6996     6987     6997     6997     6996
```

The hurdle's fitted zeros equal the observed 6996 exactly, Corollary 8.29; its AIC is the worst, 232

above the zero-inflated model's, because a zero probability without covariates or exposure ignores that a policy in force for a month has far fewer claims than one in force for a year. The zero-inflated Poisson places $\hat{\pi} = 0.279$ of the policies in the point mass and has AIC within 0.7 of the negative binomial.

6 Generalised Linear Models

FREES §13.1–13.3 (and §13.9, beyond the syllabus); ISLR §4.6.3 (PA reading list, not SRM). SRM learning outcomes 2(a), 2(b), 2(e).

6.1 The linear exponential family

Definition 8.32 (Linear exponential family; FREES (13.2))**.** Fix $\phi > 0$, a σ-finite measure λ on $\mathbb{R}$ and a function $S(\cdot, \phi)$ such that $\nu(\mathrm{d}y) = e^{S(y,\phi)}\lambda(\mathrm{d}y)$ is not concentrated at one point. Let Θ be the set of θ with $A(\theta) = \int e^{y\theta/\phi}\,\nu(\mathrm{d}y) < \infty$, assume its interior Θ° is non-empty, and put

$b(\theta) = \phi \ln A(\theta)$. The **linear exponential family** is the set of densities with respect to λ

$$f(y;\theta,\phi) = \exp\left\{\frac{y\theta - b(\theta)}{\phi} + S(y,\phi)\right\}, \qquad \theta \in \Theta^\circ, \tag{12}$$

with **natural parameter** θ and **scale parameter** ϕ. Defining b from A makes $\int f \,\mathrm{d}\lambda = 1$ hold by construction.

Theorem 8.33 (Moments; FREES §13.2.1 and, beyond the syllabus, §13.9.2)**.** b has derivatives of every order on Θ°, and for y with density (12)

$$\mathsf{E}\, y = b'(\theta), \qquad \operatorname{var} y = \phi\, b''(\theta) > 0. \tag{13}$$

So b is strictly convex (the prerequisites' lemma **convexity from the derivative**) and b' strictly increasing on Θ°.

Proof. Fix $\theta_0 \in \Theta^\circ$ and $\delta > 0$ with $[\theta_0 - 2\delta, \theta_0 + 2\delta] \subset \Theta$.
Domination. For $k \geqslant 0$ and $u \geqslant 0$, $e^u \geqslant u^k/k!$, so $|y|^k \leqslant k!(\phi/\delta)^k e^{|y|\delta/\phi}$. For $|\theta - \theta_0| < \delta$ the

exponent of $e^{|y|\delta/\phi}e^{y\theta/\phi}$ is at most $y(\theta_0+2\delta)/\phi$ when $y \geqslant 0$ and at most $y(\theta_0-2\delta)/\phi$ when $y < 0$. Hence

$$|y|^k e^{y\theta/\phi} \leqslant k!(\phi/\delta)^k\left\{e^{y(\theta_0+2\delta)/\phi} + e^{y(\theta_0-2\delta)/\phi}\right\} =: g_k(y),$$

ν-integrable because $\theta_0 \pm 2\delta \in \Theta$.

Differentiation. The kth θ-derivative of $e^{y\theta/\phi}$ is $(y/\phi)^k e^{y\theta/\phi}$, dominated by g_k/ϕ^k on $(\theta_0-\delta, \theta_0+\delta)$. By induction on k with the prerequisites' theorem **differentiation under the integral sign**, A has derivatives of every order there, $A^{(k)}(\theta) = \phi^{-k}\int y^k e^{y\theta/\phi}\,\nu(\mathrm{d}y)$, each continuous because differentiable. As $A > 0$, $b = \phi \ln A$ has derivatives of every order. Since $f = e^{y\theta/\phi}e^S/A$, $A' = A\,\mathsf{E}\,y/\phi$ and $A'' = A\,\mathsf{E}\,y^2/\phi^2$, so

$$b' = \phi\frac{A'}{A} = \mathsf{E}\,y, \qquad b'' = \phi\,\frac{A''A - (A')^2}{A^2} = \frac{\mathsf{E}\,y^2 - (\mathsf{E}\,y)^2}{\phi}.$$

The law of y is not concentrated at a point, so $\operatorname{var} y > 0$. □

Definition 8.34 (Variance function, weights; FREES §13.1–13.2)**.** By Theorem 8.33, $\theta \mapsto \mu = b'(\theta)$ is a strictly increasing bijection of Θ° onto its range, and

$$\operatorname{var} y = \phi\, v(\mu), \qquad v(\mu) = b''\big((b')^{-1}(\mu)\big), \tag{14}$$

the **variance function**. Observation i with a known **weight** $w_i > 0$ has scale $\phi_i = \phi/w_i$, so $\operatorname{var} y_i = \phi v(\mu_i)/w_i$.

Proposition 8.35 (Members; FREES Tables 13.1, 13.2, 13.8)**.** With λ Lebesgue measure on $\mathbb{R}$ or $(0,\infty)$, or counting measure on the support, each law below is (12) with the components shown.

Law	θ	ϕ	$b(\theta)$	$S(y,\phi)$	$v(\mu)$
$N(\mu,\sigma^2)$	μ	σ^2	$\theta^2/2$	$-\frac{y^2}{2\phi}-\frac{1}{2}\ln(2\pi\phi)$	1
Binomial(m,π)	$\operatorname{logit}\pi$	1	$m\ln(1+e^\theta)$	$\ln\binom{m}{y}$	$\mu(1-\mu/m)$
Pois(μ)	$\ln\mu$	1	e^θ	$-\ln y!$	μ
NB(r,p), r known	$\ln(1-p)$	1	$-r\ln(1-e^\theta)$	$\ln\frac{\Gamma(y+r)}{y!\Gamma(r)}$	$\mu+\mu^2/r$
Gamma(α,γ)	$-\gamma/\alpha$	$1/\alpha$	$-\ln(-\theta)$	$-\frac{\ln\phi}{\phi}-\ln\Gamma(\frac{1}{\phi})+(\frac{1}{\phi}-1)\ln y$	μ^2
IG(μ,λ_0)	$-\frac{1}{2\mu^2}$	$1/\lambda_0$	$-\sqrt{-2\theta}$	$-\frac{1}{2\phi y}-\frac{1}{2}\ln(2\pi\phi y^3)$	μ^3

Here the gamma density is $\gamma^\alpha y^{\alpha-1}e^{-\gamma y}/\Gamma(\alpha)$ and the inverse Gaussian density is $\sqrt{\lambda_0/(2\pi y^3)}\exp\{-\lambda_0(y-\mu)^2/(2\mu^2 y)\}$.

Proof. In each case expand $\ln f$ and match. Normal: $-(y-\mu)^2/(2\sigma^2)=(y\mu-\mu^2/2)/\sigma^2-y^2/(2\sigma^2)$. Binomial: $y\ln\pi+(m-y)\ln(1-\pi)=y\operatorname{logit}\pi+m\ln(1-\pi)$ and $\ln(1-\pi)=-\ln(1+e^\theta)$. Poisson: $y\ln\mu-\mu$. Negative binomial: $y\ln(1-p)+r\ln p$ with $p=1-e^\theta$. Gamma: with $\theta=-\gamma/\alpha$ and $\phi=1/\alpha$, $(y\theta-b)/\phi=-\gamma y+\alpha\ln\gamma-\alpha\ln\alpha$, and $\ln f$ minus this is $(\alpha-1)\ln y-\ln\Gamma(\alpha)+\alpha\ln\alpha$, which is the stated S. Inverse Gaussian: $-\lambda_0(y-\mu)^2/(2\mu^2 y)=\lambda_0\{-y/(2\mu^2)+1/\mu\}-\lambda_0/(2y)$, and $(y\theta-b)/\phi$ is the braced part times λ_0. In each row $v(\mu)$ is b'' at $\theta=(b')^{-1}(\mu)$ by Theorem 8.33:

for the gamma $b' = -1/\theta = \mu$ and $b'' = 1/\theta^2 = \mu^2$; for the inverse Gaussian $b' = (-2\theta)^{-1/2} = \mu$ and $b'' = (-2\theta)^{-3/2} = \mu^3$; for the negative binomial $b' = re^\theta/(1-e^\theta) = \mu$ and $b'' = b'/(1-e^\theta) = \mu(1+\mu/r)$. □

Remark (FREES Table 13.8, §13.1)**.** FREES's gamma entry has $S(y,\phi) = +\phi^{-1}\ln\phi - \cdots$; the sign is negative, and FREES's version multiplies every density by $e^{2\phi^{-1}\ln\phi} = \phi^{2/\phi}$, which is $0.4^5 = 0.010$ at $\alpha = 2.5$. Its inverse Gaussian entry has $\theta/(\phi y)$ in place of $-1/(2\phi y)$, a term that depends on μ. In §13.1, FREES writes $\mathbf{x}_i'\boldsymbol{\beta} = \exp(\mu_i)/(1+\exp(\mu_i))$ for logistic regression; the systematic component is $\mathbf{x}_i'\boldsymbol{\beta} = \operatorname{logit}\mu_i$. ISLR §4.6.3 lists the negative binomial among exponential families, which holds with r known, as in the table.

6.2 Link functions

Definition 8.36 (Generalised linear model; FREES (13.1), ISLR (4.42))**.** A **generalised linear model** has independent $y_1, \ldots, y_n$ from one linear exponential family with weights w_i and natural parameters θ_i, a strictly monotone differentiable **link** g with $g' \neq 0$, and the **systematic**

component

$$\eta_i = \mathbf{x}_i'\boldsymbol{\beta} = g(\mu_i), \qquad \mu_i = b'(\theta_i). \tag{15}$$

The link is **canonical** when $g = (b')^{-1}$, that is when $\theta_i = \eta_i$. ISLR's linear, logistic and Poisson regressions are the normal, Bernoulli and Poisson members with their canonical links μ, $\operatorname{logit}\mu$ and $\ln\mu$.

Remark (Links for the Bernoulli; FREES §13.2.2)**.** A Bernoulli mean lies in $(0,1)$, and the logit $\ln\{\mu/(1-\mu)\}$, the probit $\Phi^{-1}(\mu)$ and the complementary log-log $\ln\{-\ln(1-\mu)\}$ map $(0,1)$ onto $\mathbb{R}$, so every $\mathbf{x}_i'\boldsymbol{\beta}$ gives a mean in $(0,1)$; the identity and the logarithm do not. Under the logarithmic link, Proposition 8.10(iv) gives a multiplicative rating plan $\mu = B \cdot A_i \cdot T_j$; under the identity link the plan is additive. A link is often written up to a constant factor ($1/\mu^2$ for the inverse Gaussian), which rescales $\boldsymbol{\beta}$.

6.3 Estimation

Proposition 8.37 (Canonical links; FREES §13.3.1)**.** Under the canonical link,

$$L(\boldsymbol{\beta},\phi)=\sum_i\left\{\frac{w_i\{y_i\mathbf{x}_i'\boldsymbol{\beta}-b(\mathbf{x}_i'\boldsymbol{\beta})\}}{\phi}+S(y_i,\phi/w_i)\right\},$$

the score is $\phi^{-1}\sum_i w_i(y_i-\mu_i)\mathbf{x}_i$, and $-\nabla^2 L=\phi^{-1}\sum_i w_i b''(\mathbf{x}_i'\boldsymbol{\beta})\mathbf{x}_i\mathbf{x}_i'=\mathcal{I}(\boldsymbol{\beta})$ whatever the responses. L is concave in $\boldsymbol{\beta}$, strictly when $\operatorname{rank}\mathbf{X}=k+1$, and the maximiser solves the normal equations $\sum_i w_i(y_i-\mu_i)\mathbf{x}_i=\mathbf{0}$, which do not involve ϕ.

Proof. Substitute $\theta_i=\mathbf{x}_i'\boldsymbol{\beta}$ and $\phi_i=\phi/w_i$ into (12). Differentiating, $\nabla L=\phi^{-1}\sum_i w_i\{y_i-b'(\mathbf{x}_i'\boldsymbol{\beta})\}\mathbf{x}_i$ and $\nabla^2 L=-\phi^{-1}\sum_i w_i b''(\mathbf{x}_i'\boldsymbol{\beta})\mathbf{x}_i\mathbf{x}_i'$, free of the y_i, hence equal to its expectation, which is $-\operatorname{var}\nabla L$ by Definition 8.34. $b''>0$ (Theorem 8.33) makes $-\nabla^2 L$ positive semi-definite, and positive definite when the $\mathbf{x}_i$ span, which gives the concavity claims by the prerequisites' lemma **convexity from the derivative**. □

Theorem 8.38 (Score and information, general link; FREES (13.13)–(13.14))**.** For any link,

$$\nabla L(\boldsymbol{\beta}) = \sum_{i=1}^{n} \frac{\partial \mu_i}{\partial \boldsymbol{\beta}} \frac{y_i - \mu_i}{\operatorname{var} y_i} = \sum_{i=1}^{n} \frac{w_i}{\phi} \frac{y_i - \mu_i}{v(\mu_i)} \frac{\mathrm{d}\mu_i}{\mathrm{d}\eta_i} \mathbf{x}_i, \qquad \mathcal{I}(\boldsymbol{\beta}) = \sum_{i=1}^{n} \frac{w_i}{\phi\, v(\mu_i)} \left(\frac{\mathrm{d}\mu_i}{\mathrm{d}\eta_i}\right)^2 \mathbf{x}_i \mathbf{x}_i'.$$

Proof. $\partial L/\partial \theta_i = w_i(y_i - \mu_i)/\phi$, since $b'(\theta_i) = \mu_i$. Differentiating $\mu_i = b'(\theta_i)$ gives $\mathrm{d}\mu_i/\mathrm{d}\theta_i = b''(\theta_i) = v(\mu_i)$, so by the chain rule through $\boldsymbol{\beta} \mapsto \eta_i \mapsto \mu_i \mapsto \theta_i$,

$$\frac{\partial L}{\partial \boldsymbol{\beta}} = \sum_i \frac{w_i(y_i - \mu_i)}{\phi} \frac{1}{v(\mu_i)} \frac{\mathrm{d}\mu_i}{\mathrm{d}\eta_i} \mathbf{x}_i,$$

which is both displayed forms, as $\operatorname{var} y_i = \phi v(\mu_i)/w_i$ and $\partial \mu_i/\partial \boldsymbol{\beta} = (\mathrm{d}\mu_i/\mathrm{d}\eta_i)\mathbf{x}_i$. The summands are independent with mean zero, so $\mathcal{I} = \operatorname{var} \nabla L$ is the sum of $(w_i/\phi)^2 v(\mu_i)^{-2} (\mathrm{d}\mu_i/\mathrm{d}\eta_i)^2 \operatorname{var} y_i\, \mathbf{x}_i \mathbf{x}_i'$; substitute $\operatorname{var} y_i$. □

Corollary 8.39 (No covariates; FREES §13.9.3 (beyond the syllabus))**.** If $\mathbf{x}_i = 1$ for all i, the maximum likelihood estimate of μ is the weighted mean $\bar{y}_w = \sum_i w_i y_i / \sum_i w_i$, whatever the family

and the link, provided $\bar{y}_w$ lies in the range of b'.

Proof. The score is $(\mathrm{d}\mu/\mathrm{d}\eta)\{\phi v(\mu)\}^{-1}\sum_i w_i(y_i-\mu)$, with a nonzero factor in front, and $\sum_i w_i(y_i-\mu)$ is strictly decreasing in μ and zero at $\bar{y}_w$. Since μ is a strictly monotone function of β, L increases in μ below $\bar{y}_w$ and decreases above it. □

Theorem 8.40 (Iteratively reweighted least squares; FREES §13.9.4 (beyond the syllabus))**.** At the current $\boldsymbol{\beta}^{(m)}$ let

$$W_i = \frac{w_i}{\phi\, v(\mu_i)}\left(\frac{\mathrm{d}\mu_i}{\mathrm{d}\eta_i}\right)^2, \qquad z_i = \eta_i + \frac{y_i-\mu_i}{\mathrm{d}\mu_i/\mathrm{d}\eta_i}.$$

The Fisher scoring step $\boldsymbol{\beta}^{(m+1)} = \boldsymbol{\beta}^{(m)} + \mathcal{I}(\boldsymbol{\beta}^{(m)})^{-1}\nabla L(\boldsymbol{\beta}^{(m)})$ equals the weighted least squares fit $(\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\mathbf{z}$ with $\mathbf{W} = \mathrm{diag}(W_i)$, and does not involve ϕ. Under the canonical link it is the Newton step.

Proof. By Theorem 8.38, $\mathcal{I} = \mathbf{X}'\mathbf{W}\mathbf{X}$ and $\nabla L = \sum_i W_i(z_i-\eta_i)\mathbf{x}_i = \mathbf{X}'\mathbf{W}(\mathbf{z}-\mathbf{X}\boldsymbol{\beta}^{(m)})$, because $W_i(z_i-\eta_i) = W_i(y_i-\mu_i)/(\mathrm{d}\mu_i/\mathrm{d}\eta_i)$ is the ith score coefficient. So $\boldsymbol{\beta}^{(m)} + (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}(\mathbf{z}-$

$\mathbf{X}\boldsymbol{\beta}^{(m)}) = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\mathbf{z}$, and the factor $1/\phi$ common to $\mathbf{W}$ cancels. Under the canonical link $\mathcal{I} = -\nabla^2 L$ (Proposition 8.37). □

Remark (FREES (13.15) (beyond the syllabus))**.** FREES's Newton–Raphson step (13.15) subtracts $(\sum_i w_i b'' \mathbf{x}_i \mathbf{x}_i')^{-1} \sum_i w_i (y_i - b')\mathbf{x}_i$ from $\boldsymbol{\beta}_{\text{OLD}}$; the step adds it, as the Hessian is $-\phi^{-1}\sum_i w_i b'' \mathbf{x}_i \mathbf{x}_i'$. Its final weighted least squares formula is right.

Theorem 8.41 (Large-sample inference, canonical link; FREES §13.3.1)**.** Consider a family with $\Theta^\circ = \mathbb{R}$ (normal, binomial, Poisson) under its canonical link, and let $\mathbf{b}$ maximise $\ell_n(\boldsymbol{\beta}) = \sum_i w_i\{y_i \mathbf{x}_i'\boldsymbol{\beta} - b(\mathbf{x}_i'\boldsymbol{\beta})\}$. Suppose the y_i are independent with $\mathsf{E}\, y_i = b'(\mathbf{x}_i'\boldsymbol{\beta}_0)$ and $\operatorname{var} y_i = \tau_i^2$, that $|\mathbf{x}_i| \leqslant M$, $c_w \leqslant w_i \leqslant C_w$ and $\sup_i \mathsf{E}(y_i - \mathsf{E}\, y_i)^4 < \infty$, and that

$$\mathbf{G}_n = \frac{1}{n}\sum_i w_i b''(\mathbf{x}_i'\boldsymbol{\beta}_0)\mathbf{x}_i\mathbf{x}_i' \to \mathbf{G}, \qquad \mathbf{F}_n = \frac{1}{n}\sum_i w_i^2 \tau_i^2 \mathbf{x}_i\mathbf{x}_i' \to \mathbf{F},$$

both positive definite. Then with probability tending to one $\mathbf{b}$ exists and is unique, $\mathbf{b} \xrightarrow{p} \boldsymbol{\beta}_0$, and $\sqrt{n}(\mathbf{b} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{G}^{-1}\mathbf{F}\mathbf{G}^{-1})$. If $\operatorname{var} y_i = \phi\, v(\mu_i)/w_i$, then $\mathbf{F} = \phi\mathbf{G}$ and the limit is

$N(\mathbf{0}, \phi\mathbf{G}^{-1})$.

Proof. Apply the prerequisites' theorem **concave maximum likelihood** with $\ell_i = w_i\{y_i\mathbf{x}_i'\boldsymbol{\beta} - b(\mathbf{x}_i'\boldsymbol{\beta})\}$, concave and smooth by Theorem 8.33. Its score $w_i(y_i - \mu_i)\mathbf{x}_i$ has mean $\mathbf{0}$, average variance $\mathbf{F}_n \to \mathbf{F}$, and fourth moments at most $C_w^4 M^4 \sup_i \mathsf{E}(y_i - \mu_i)^4$, so the Lyapunov corollary with $\delta = 2$ gives the normal limit of the score. The Hessian $\mathbf{H}_n(\boldsymbol{\beta}) = \sum_i w_i b''(\mathbf{x}_i'\boldsymbol{\beta})\mathbf{x}_i\mathbf{x}_i'$ is non-random with $n^{-1}\mathbf{H}_n(\boldsymbol{\beta}_0) = \mathbf{G}_n$. For $|\boldsymbol{\beta} - \boldsymbol{\beta}_0| \leqslant 1$ the arguments $\mathbf{x}_i'\boldsymbol{\beta}$ lie in $J = [-M(|\boldsymbol{\beta}_0| + 1), M(|\boldsymbol{\beta}_0| + 1)]$, where the continuous b''' is bounded by some K (the prerequisites' theorem **uniform continuity on compacts**); the mean value theorem gives $\|\mathbf{H}_n(\boldsymbol{\beta}) - \mathbf{H}_n(\boldsymbol{\beta}_0)\| \leqslant nC_wKM^3|\boldsymbol{\beta} - \boldsymbol{\beta}_0|$. Parts (i)–(iii) of the theorem give the conclusions, with uniqueness from strict concavity once $\operatorname{rank}\mathbf{X} = k + 1$, which $\mathbf{G}$ positive definite forces for large n (two maximisers would give a strictly larger value at their midpoint). If $\operatorname{var} y_i = \phi v(\mu_i)/w_i = \phi b''(\mathbf{x}_i'\boldsymbol{\beta}_0)/w_i$, then $w_i^2\tau_i^2 = \phi w_i b''(\mathbf{x}_i'\boldsymbol{\beta}_0)$ and $\mathbf{F}_n = \phi\mathbf{G}_n$. □

Remark (FREES §13.1, §13.3.1–13.3.2)**.** Theorem 8.41 is FREES's statement that inference "only requires that we choose the mean and variance functions correctly": consistency uses only the mean,

and the model-based variance $\phi\mathbf{G}^{-1}$ also the variance function, the rest of the distribution being irrelevant. When $\operatorname{var} y_i = \sigma^2\phi v(\mu_i)/w_i$ with an extra factor σ^2 (FREES §13.3.2), $\mathbf{F} = \sigma^2\phi\mathbf{G}$ and the standard errors scale by σ, estimated by the Pearson statistic $\sum_i w_i(y_i-\hat\mu_i)^2/\{\phi v(\hat\mu_i)\}$ over $n-(k+1)$, an estimate proved consistent here for the Poisson family without weights (Proposition 8.17). FREES's "empirical" standard error in §13.9.3 (beyond the syllabus) inverts $\sum_i(\partial\mu_i/\partial\boldsymbol\beta)(y_i-\mu_i)^{-2}(\partial\mu_i/\partial\boldsymbol\beta')$; the estimate of $\mathbf{G}^{-1}\mathbf{F}\mathbf{G}^{-1}$ is the sandwich $\hat{\mathbf{G}}^{-1}\hat{\mathbf{F}}\hat{\mathbf{G}}^{-1}$ with $(y_i-\hat\mu_i)^2$ in $\hat{\mathbf{F}}$, as in Proposition 8.20, which proves its consistency for the Poisson family.

Example 8.12 (Ratemaking classification; FREES §13.2.2–13.3.1, Tables 13.3–13.4)**.** FREES's `AutoCollision`: average severity of 8942 UK collision claims in 32 cells of eight driver age groups by four vehicle uses, fitted by a gamma model with logarithmic link and the claim counts as weights w_i, so that $\operatorname{var} y_i = \phi\mu_i^2/w_i$ for a cell average of w_i claims:

```
AC <- read.csv("data/frees/csv/AutoCollision.csv", quote = "",
               fileEncoding = "latin1"); AC$Age <- sub("\u0096", "-", AC$Age)
gA <- glm(Severity ~ factor(Age) + factor(Vehicle_Use), weights = Claim_Count,
          data = AC, family = Gamma(link = "log"))
cA <- coef(gA); levels(factor(AC$Vehicle_Use))
```

```
## [1] "Business"   "DriveLong"  "DriveShort" "Pleasure"
```

```
prem <- exp(cA[[1]]) * outer(exp(c(0, cA[2:8])), exp(c(cA[9:11], 0)))
dimnames(prem) <- list(levels(factor(AC$Age)), c("DriveLong", "DriveShort",
                                                 "Pleasure", "Business"))
print(round(cbind(prem, Average = rowMeans(prem)), 2), digits = 7)
```

```
##       DriveLong DriveShort Pleasure Business Average
## 17-20    322.17     265.56   254.90   419.07  315.42
## 21-24    320.66     264.31   253.70   417.10  313.94
## 25-29    297.26     245.02   235.19   386.66  291.03
## 30-34    284.85     234.80   225.37   370.53  278.89
## 35-39    229.37     189.06   181.47   298.35  224.56
## 40-49    248.15     204.54   196.33   322.78  242.95
## 50-59    251.95     207.67   199.34   327.72  246.67
## 60+      246.47     203.16   195.00   320.60  241.31
```

```
print(round(c(colMeans(prem), overall = mean(prem)), 2), digits = 7)
```

```
##  DriveLong DriveShort   Pleasure   Business    overall
##     275.11     226.77     217.66     357.85     269.35
```

FREES's Table 13.4 is reproduced to the cent. Its first two columns are labelled "DriverShort" and

"DriveLong" in that order; the script takes the coefficients in R's alphabetical order, Business (the base), DriveLong, DriveShort, Pleasure, so the column 322.17, ... is DriveLong and 265.56, ... is DriveShort, matching the raw averages of Table 13.3, where long commutes have the higher severities.

6.4 Goodness of fit

Proposition 8.42 (The cross-product term; FREES §13.3.3)**.** With $\hat{y}_i$ any fitted values, $\sum_i (y_i - \bar{y})^2 = \sum_i (y_i - \hat{y}_i)^2 + \sum_i (\hat{y}_i - \bar{y})^2 + 2\sum_i (y_i - \hat{y}_i)(\hat{y}_i - \bar{y})$. The last sum is zero for least squares with an intercept; for a canonical-link generalised linear model with an intercept and $w_i = 1$ it equals $\sum_i (y_i - \hat{\mu}_i)\hat{\mu}_i$, which need not vanish.

Proof. Expand $\{(y_i - \hat{y}_i) + (\hat{y}_i - \bar{y})\}^2$. For least squares, $\mathbf{X}'\mathbf{e} = \mathbf{0}$ makes $\mathbf{e}$ orthogonal to $\hat{\mathbf{Y}}$ and to $\mathbf{1}$. Under a canonical link $\sum_i (y_i - \hat{\mu}_i) = 0$ by Proposition 8.37, which removes $\bar{y}$; the normal equations make $\mathbf{Y} - \hat{\boldsymbol{\mu}}$ orthogonal to $\mathbf{Xb}$, the vector of $\hat{\eta}_i$, not to the vector of $\hat{\mu}_i = b'(\hat{\eta}_i)$. □

Definition 8.43 (Deviance; FREES §13.3.3)**.** Let $\tilde{\theta}_i$ solve $b'(\tilde{\theta}_i) = y_i$ (the **saturated** fit) and let $\hat{\theta}_i$ be the fitted natural parameters. The **scaled deviance** is $D^* = 2\{L(\tilde{\boldsymbol{\theta}}) - L(\hat{\boldsymbol{\theta}})\}$ and the **deviance** is $D = \phi D^*$:

$$D = 2\sum_{i=1}^{n} w_i\big\{y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i)\big\} = \sum_{i=1}^{n} d_i. \tag{16}$$

Proposition 8.44 (Non-negativity and special cases; FREES §13.3.3)**.** Every $d_i \geqslant 0$, with equality only if $\hat{\mu}_i = y_i$. With $w_i = 1$ and the convention $0 \ln 0 = 0$,

$$\text{normal:} \quad D = \sum_i (y_i - \hat{\mu}_i)^2,$$

$$\text{Bernoulli:} \quad D = 2\sum_i \left\{ y_i \ln \frac{y_i}{\hat{\mu}_i} + (1 - y_i) \ln \frac{1 - y_i}{1 - \hat{\mu}_i} \right\},$$

$$\text{Poisson:}\quad D = 2\sum_i \Big\{ y_i \ln \frac{y_i}{\hat\mu_i} - (y_i - \hat\mu_i) \Big\},$$

$$\text{gamma:}\quad D = 2\sum_i \Big\{ -\ln \frac{y_i}{\hat\mu_i} + \frac{y_i - \hat\mu_i}{\hat\mu_i} \Big\}.$$

For y_i at the boundary of the range of b' (0 or 1 for the Bernoulli, 0 for the Poisson), d_i is defined as the limit as $\tilde\theta_i \to \pm\infty$, which the convention gives.

Proof. With $h(\theta) = y_i\theta - b(\theta)$, $d_i = 2w_i\{h(\tilde\theta_i) - h(\hat\theta_i)\}$; $h' = y_i - b'$ vanishes at $\tilde\theta_i$ and $h'' = -b'' < 0$, so h is strictly concave (the prerequisites' lemma **convexity from the derivative**) with its maximum at $\tilde\theta_i$. Normal: $b = \theta^2/2$, $\theta = \mu$, so $d_i = 2\{y_i(y_i - \hat\mu_i) - y_i^2/2 + \hat\mu_i^2/2\} = (y_i - \hat\mu_i)^2$. Poisson: $b = e^\theta$, $\theta = \ln\mu$, so $d_i = 2\{y_i \ln y_i - y_i \ln\hat\mu_i - y_i + \hat\mu_i\}$; at $y_i = 0$, $h(\theta) = -e^\theta \uparrow 0$ as $\theta \to -\infty$, so $d_i = 2\hat\mu_i$, the formula's value. Bernoulli: $b = \ln(1 + e^\theta)$, $h(\theta) = y_i \ln\mu + (1 - y_i)\ln(1 - \mu)$ with $\mu = b'(\theta)$, whose supremum over μ is $y_i \ln y_i + (1 - y_i)\ln(1 - y_i)$, zero for $y_i \in \{0, 1\}$. Gamma: $b = -\ln(-\theta)$, $\theta = -1/\mu$, so $d_i = 2\{-1 + y_i/\hat\mu_i - \ln y_i + \ln\hat\mu_i\}$. □

Remark (FREES §13.3.3)**.** FREES's Bernoulli and Poisson deviances lack the factor 2 of $D = \phi D^*$, and its Poisson term has $+(y_i - \hat{\mu}_i)$; with an intercept and the canonical link $\sum_i (y_i - \hat{\mu}_i) = 0$, so that sign does not change D.

Example 8.13 (The cross-product on `Bikeshare`; FREES §13.3.3)**.**

```
yB <- B$bikers; mB <- fitted(mpo)
round(c(cross_product = sum((yB - mB) * mB), total_SS = sum((yB - mean(yB))^2),
        R2_one_minus_error = 1 - sum((yB - mB)^2) / sum((yB - mean(yB))^2),
        R2_regression = sum((mB - mean(yB))^2) / sum((yB - mean(yB))^2)), 4)
```

```
##       cross_product            total_SS R2_one_minus_error
##        -777481.1582     154743727.7335              0.7338
##       R2_regression
##              0.7439
```

The cross-product sum is -7.77×10^5, so the two definitions of R^2 differ: 0.7338 and 0.7439.

7 Application: Medical Expenditures

FREES §13.4. SRM learning outcomes 2(c), 2(d), 2(e).

Proposition 8.45 (Gamma regression with logarithmic link)**.** Let $\ln \mu_i = \mathbf{x}_i'\boldsymbol{\beta}$ and let y_i be gamma with mean μ_i and shape ν, so $\operatorname{var} y_i = \mu_i^2/\nu$. Then, with $\ell_n(\boldsymbol{\beta}) = -\sum_i (y_i/\mu_i + \ln \mu_i)$,

(i) $L(\boldsymbol{\beta}, \nu) = \nu \ell_n(\boldsymbol{\beta}) + c(\nu, \mathbf{Y})$, so the maximiser $\mathbf{b}$ does not depend on ν;

(ii) ℓ_n is concave, strictly when $\operatorname{rank} \mathbf{X} = k+1$, with $\nabla \ell_n = \sum_i (y_i/\mu_i - 1)\mathbf{x}_i$ and $-\nabla^2 \ell_n = \sum_i (y_i/\mu_i)\mathbf{x}_i\mathbf{x}_i'$;

(iii) the observed information for $\boldsymbol{\beta}$ is $\nu \sum_i (y_i/\mu_i)\mathbf{x}_i\mathbf{x}_i'$ and the expected information is $\nu \mathbf{X}'\mathbf{X}$; they differ because the link is not canonical.

Proof. The gamma density with shape ν and rate ν/μ has $\ln f = \nu \ln \nu - \nu \ln \mu + (\nu - 1)\ln y - \nu y/\mu - \ln \Gamma(\nu)$, which is (i). With $\mu_i = e^{\eta_i}$, $-(y_i e^{-\eta} + \eta)$ has derivatives $y_i e^{-\eta} - 1$ and $-y_i e^{-\eta} < 0$ in η, which with $\partial \eta_i / \partial \boldsymbol{\beta} = \mathbf{x}_i$ gives (ii) by the prerequisites' lemma **convexity from the derivative**; the Hessian is negative definite when the $\mathbf{x}_i$ span, as every $y_i > 0$. Multiplying by ν gives the observed information, and $\mathsf{E}\, y_i/\mu_i = 1$ the expected one. The canonical link is $-1/\mu$

(Proposition 8.35). □

Theorem 8.46 (Large-sample inference for gamma regression)**.** Let the $y_i > 0$ be independent with $\mathsf{E}\, y_i = \mu_i = e^{\mathbf{x}_i'\boldsymbol{\beta}_0}$ and $\operatorname{var} y_i = \phi\mu_i^2$, let $|\mathbf{x}_i| \leqslant M$, $\sup_i \mathsf{E}(y_i - \mu_i)^4 < \infty$, and $n^{-1}\mathbf{X}'\mathbf{X} \to \mathbf{G}$ positive definite. Then $\mathbf{b}$ of Proposition 8.45 exists and is unique with probability tending to one, $\mathbf{b} \xrightarrow{p} \boldsymbol{\beta}_0$,

$$\sqrt{n}(\mathbf{b} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \phi\mathbf{G}^{-1}), \qquad n^{-1}\sum_i (y_i/\hat{\mu}_i)\mathbf{x}_i\mathbf{x}_i' \xrightarrow{p} \mathbf{G},$$

the Pearson estimate $\hat{\phi} = \{n - (k+1)\}^{-1}\sum_i (y_i/\hat{\mu}_i - 1)^2$ satisfies $\hat{\phi} \xrightarrow{p} \phi$, and so standard errors from the observed and from the expected information, each with a consistent estimate of ϕ, are both consistent.

Proof. Apply the prerequisites' theorem **concave maximum likelihood** to ℓ_n, concave and smooth by Proposition 8.45. The means lie in $[e^{-M|\boldsymbol{\beta}_0|}, e^{M|\boldsymbol{\beta}_0|}]$.

Score. $(y_i/\mu_i - 1)\mathbf{x}_i$ has mean $\mathbf{0}$, variance $\phi\mathbf{x}_i\mathbf{x}_i'$, and fourth moments at most $M^4 e^{4M|\boldsymbol{\beta}_0|}\sup_i \mathsf{E}(y_i - \mu_i)^4$; so $\mathbf{F}_n = \phi n^{-1}\mathbf{X}'\mathbf{X} \to \phi\mathbf{G}$ and the Lyapunov corollary applies with

fourth moments.

Curvature. $\mathbf{H}_n(\boldsymbol{\beta}) = \sum_i y_i e^{-\mathbf{x}_i'\boldsymbol{\beta}} \mathbf{x}_i \mathbf{x}_i'$. Each entry of $n^{-1}\mathbf{H}_n(\boldsymbol{\beta}_0) - n^{-1}\mathbf{X}'\mathbf{X}$ is an average of independent mean-zero terms $(y_i/\mu_i - 1)x_{ij}x_{il}$ of variance at most ϕM^4, so it tends to 0 by Chebyshev's inequality, and $\mathbf{G}$ is the limit. For $|\boldsymbol{\beta} - \boldsymbol{\beta}_0| \leqslant 1$, with $K = e^{M(|\boldsymbol{\beta}_0|+1)}$, the mean value theorem gives $|e^{-\mathbf{x}_i'\boldsymbol{\beta}} - e^{-\mathbf{x}_i'\boldsymbol{\beta}_0}| \leqslant KM|\boldsymbol{\beta} - \boldsymbol{\beta}_0|$, so $\|\mathbf{H}_n(\boldsymbol{\beta}) - \mathbf{H}_n(\boldsymbol{\beta}_0)\| \leqslant nC_n|\boldsymbol{\beta} - \boldsymbol{\beta}_0|$ with $C_n = KM^3 n^{-1}\sum_i y_i$. $\mathsf{E}\, C_n \leqslant KM^3 e^{M|\boldsymbol{\beta}_0|}$, so $\mathsf{P}(C_n > c) \leqslant \mathsf{E}\, C_n/c$ by Markov's inequality and C_n is bounded in probability. This is the curvature condition.

Parts (i), (ii), (iii) and (v) of the theorem give existence, uniqueness (by Proposition 8.45(ii) ℓ_n is strictly concave once $\operatorname{rank}\mathbf{X} = k+1$, which $n^{-1}\mathbf{X}'\mathbf{X} \to \mathbf{G}$ positive definite forces for large n, and two maximisers would give a strictly larger value at their midpoint), consistency, the sandwich $\mathbf{G}^{-1}(\phi\mathbf{G})\mathbf{G}^{-1} = \phi\mathbf{G}^{-1}$, and the limit of the observed information; $n^{-1}\mathbf{X}'\mathbf{X} \to \mathbf{G}$ by assumption.

Dispersion. Follow the proof of Proposition 8.17 with μ_i^2 in place of μ_i. Write $e_i = y_i - \mu_i$, $\delta_i = \hat{\mu}_i - \mu_i$, $\Delta = \max_i |\delta_i|$, $m_- = e^{-M|\boldsymbol{\beta}_0|}$ and $m_+ = e^{M|\boldsymbol{\beta}_0|}$. On $|\mathbf{b} - \boldsymbol{\beta}_0| \leqslant 1$ the mean value theorem gives $\Delta \leqslant KM|\mathbf{b} - \boldsymbol{\beta}_0|$, so $\Delta \to 0$ in probability. The average $n^{-1}\sum_i e_i^2/\mu_i^2$ has mean ϕ and variance at most $\sup_i \mathsf{E}\, e_i^4/(nm_-^4)$, so it tends to ϕ in probability by Chebyshev's inequality.

On $\{\Delta \leqslant m_-/2\}$, every $\hat{\mu}_i \geqslant m_-/2$, and

$$\frac{(e_i - \delta_i)^2}{\hat{\mu}_i^2} - \frac{e_i^2}{\mu_i^2} = \frac{e_i^2(\mu_i^2 - \hat{\mu}_i^2)}{\mu_i^2 \hat{\mu}_i^2} + \frac{\delta_i^2 - 2e_i\delta_i}{\hat{\mu}_i^2}$$

with $|\mu_i^2 - \hat{\mu}_i^2| \leqslant \Delta(2m_+ + \Delta)$. The average of these differences is therefore at most $4\Delta(2m_+ + \Delta)m_-^{-4}n^{-1}\sum_i e_i^2 + 4m_-^{-2}(\Delta^2 + 2\Delta n^{-1}\sum_i |e_i|)$ in absolute value, which tends to 0 in probability as in Proposition 8.17; with $n/\{n-(k+1)\} \to 1$, $\hat{\phi} \xrightarrow{p} \phi$. □

Example 8.14 (Inpatient expenditures; FREES §13.4, Table 13.5)**.** FREES's MEPS sample (Unit 7) restricted to the 157 persons with positive inpatient expenditure `EXPENDIP`; race `OTHER` is merged into `WHITE`, as in Unit 7:

```
H <- read.csv("data/frees/csv/HealthExpend.csv")
D <- subset(H, EXPENDIP > 0); D$RACE[D$RACE == "OTHER"] <- "WHITE"
D$RACE <- relevel(factor(D$RACE), ref = "WHITE")
D$REGION <- relevel(factor(D$REGION), ref = "WEST")
D$EDUC <- relevel(factor(D$EDUC), ref = "LHIGHSC")
D$PHSTAT <- relevel(factor(D$PHSTAT), ref = "EXCE")
```

```r
D$INCOME <- relevel(factor(D$INCOME), ref = "HINCOME")
c(n = nrow(D), female_pct = round(100 * mean(D$GENDER), 1),
  median_female = median(D$EXPENDIP[D$GENDER == 1]),
  median_male = median(D$EXPENDIP[D$GENDER == 0]),
  median_all = median(D$EXPENDIP), maximum = max(D$EXPENDIP))
```

```
##            n    female_pct median_female   median_male
##          157            72          5546          7313
##   median_all       maximum
##         5695        607801
```

```r
c(median_COUNTIP = median(D$COUNTIP), median_AGE = median(D$AGE))
```

```
## median_COUNTIP     median_AGE
##              1             41
```

```r
do.call(rbind, lapply(c("RACE", "REGION", "EDUC", "PHSTAT", "MNHPOOR", "ANYLIMIT",
  "INCOME", "insure"),
  function(v) { g <- factor(D[[v]])
    data.frame(variable = v, level = levels(g), pct = round(100 *
      as.vector(prop.table(table(g))), 1),
                median = round(as.vector(tapply(D$EXPENDIP, g, median)))) }))
```

```
##    variable     level  pct median
## 1      RACE     WHITE 75.8   5695
## 2      RACE     ASIAN  2.5   4003
## 3      RACE     BLACK 19.7   6100
## 4      RACE     NATIV  1.9   2310
## 5    REGION      WEST 19.1   4297
## 6    REGION   MIDWEST 21.7   7999
## 7    REGION NORTHEAST 18.5   5833
## 8    REGION     SOUTH 40.8   5595
## 9      EDUC   LHIGHSC 33.1   5338
## 10     EDUC   COLLEGE 23.6   5611
## 11     EDUC   HIGHSCH 43.3   5907
## 12   PHSTAT      EXCE 16.6   5277
## 13   PHSTAT      FAIR 10.2   5228
## 14   PHSTAT      GOOD 31.2   5032
## 15   PHSTAT      POOR 17.2  10447
## 16   PHSTAT      VGOO 24.8   5546
## 17  MNHPOOR         0 84.1   5599
## 18  MNHPOOR         1 15.9   6583
## 19 ANYLIMIT         0 58.6   4746
```

```
## 20 ANYLIMIT         1 41.4   7826
## 21   INCOME   HINCOME 21.7   7271
## 22   INCOME   LINCOME 16.6   6909
## 23   INCOME   MINCOME 26.8   5851
## 24   INCOME     NPOOR  7.0   5546
## 25   INCOME      POOR 28.0   4097
## 26   insure         0  8.9   2668
## 27   insure         1 91.1   5943
```

72.0% are women, with median 5546 against 7313 for men; the overall median is 5695 and the largest expenditure 607,800.6, FREES's figures. The medians by level of every other variable are FREES's Table 13.5 to the dollar (`OTHER` merged into `WHITE`, as in FREES's table); three race percentages differ from FREES's by 0.1 to 0.2 points (FREES prints 2.6, 19.8 and 75.6 where the data give 2.5, 19.7 and 75.8), a difference the merge does not explain. The poor self-rated physical health group stands out, with median 10,447.

Example 8.15 (Gamma and inverse Gaussian fits; FREES Table 13.6)**.** By Proposition 8.45(i) the gamma **b** is `glm`'s, whatever ν, and $\hat{\nu}$ maximises the log-likelihood at it; the same holds for the inverse Gaussian with ϕ. FREES's t-values use the observed information of the joint fit in

$(\boldsymbol{\beta}, \ln \nu)$, computed numerically:

```
suppressMessages(library(numDeriv))
ctl <- glm.control(epsilon = 1e-14, maxit = 500)
fit_ml <- function(g, ldens, lo, hi) {        # b from glm; scale by a 1-d search
  X <- model.matrix(g); yy <- g$y; b <- coef(g)
  ls <- optimize(function(s) sum(ldens(yy, fitted(g), exp(s))), c(lo, hi),
                 maximum = TRUE, tol = 1e-12)$maximum
  nll <- function(th) { k <- length(th)
    -sum(ldens(yy, exp(drop(X %*% th[-k])), exp(th[k]))) }
  th <- c(b, ls); V <- solve(hessian(nll, th)); k <- length(th)
  list(b = b, t = b / sqrt(diag(V)[-k]), scale = exp(ls),
       se_logscale = sqrt(V[k, k]), logLik = -nll(th), npar = k, X = X, y = yy) }
lgam <- function(y, mu, nu) dgamma(y, shape = nu, rate = nu / mu, log = TRUE)
lig <- function(y, mu, phi) statmod::dinvgauss(y, mean = mu, dispersion = phi, log =
 ↪  TRUE)
fF <- EXPENDIP ~ COUNTIP + AGE + GENDER + RACE + REGION + EDUC + PHSTAT + MNHPOOR +
  ANYLIMIT + INCOME + insure
D$BLACK <- 1 * (D$RACE == "BLACK"); D$POOR <- 1 * (D$PHSTAT == "POOR")
D$POORNEG <- 1 * (D$INCOME == "POOR")
```

```r
fR <- EXPENDIP ~ COUNTIP + AGE + GENDER + BLACK + REGION + EDUC + POOR + MNHPOOR +
  ANYLIMIT + POORNEG
gF <- glm(fF, data = D, family = Gamma(link = "log"), control = ctl)
gR <- glm(fR, data = D, family = Gamma(link = "log"), control = ctl)
gI <- glm(fR, data = D, family = inverse.gaussian(link = "log"), start = coef(gR),
          control = ctl)
full <- fit_ml(gF, lgam, -3, 3); red <- fit_ml(gR, lgam, -3, 3)
ig <- fit_ml(gI, lig, -12, 0)
round(cbind(full_b = full$b, full_t = full$t), 3)
```

```
##                 full_b full_t
## (Intercept)      6.891 13.080
## COUNTIP          0.681  6.155
## AGE              0.021  3.024
## GENDER          -0.228 -1.263
## RACEASIAN       -0.506 -1.029
## RACEBLACK       -0.331 -1.656
## RACENATIV       -1.220 -2.217
## REGIONMIDWEST    0.255  1.062
## REGIONNORTHEAST -0.372 -1.548
```

```
## REGIONSOUTH      0.010  0.047
## EDUCCOLLEGE     -0.413 -1.723
## EDUCHIGHSCH     -0.155 -0.827
## PHSTATFAIR      -0.194 -0.641
## PHSTATGOOD       0.041  0.183
## PHSTATPOOR      -0.003 -0.010
## PHSTATVGOO       0.000  0.000
## MNHPOOR         -0.396 -1.634
## ANYLIMIT         0.010  0.053
## INCOMELINCOME    0.536  2.148
## INCOMEMINCOME    0.114  0.522
## INCOMENPOOR      0.453  1.243
## INCOMEPOOR      -0.078 -0.308
## insure           0.794  3.068
```

```
round(cbind(red_b = red$b, red_t = red$t, ig_b = ig$b, ig_t = ig$t), 3)
```

```
##                 red_b  red_t   ig_b   ig_t
## (Intercept)     7.859 17.951  6.544  3.024
## COUNTIP         0.672  5.965  1.263  0.989
## AGE             0.015  2.439  0.018  0.727
```

```
## GENDER          -0.118 -0.648  0.363  0.482
## BLACK           -0.258 -1.287 -0.321 -0.577
## REGIONMIDWEST    0.448  1.888  0.399  0.654
## REGIONNORTHEAST -0.214 -0.890  0.109  0.165
## REGIONSOUTH      0.108  0.516  0.164  0.319
## EDUCCOLLEGE     -0.469 -2.108 -0.367 -0.606
## EDUCHIGHSCH     -0.210 -1.138 -0.039 -0.078
## POOR             0.167  0.706  0.167  0.258
## MNHPOOR         -0.314 -1.337 -0.378 -0.642
## ANYLIMIT         0.052  0.266  0.218  0.287
## POORNEG         -0.406 -2.129 -0.356 -0.595
```

```r
sm <- function(m, sc) c(scale = sc, scale_t = 1 / m$se_logscale, logLik = m$logLik,
  AIC = -2 * m$logLik + 2 * m$npar, AIC_frees = -2 * m$logLik + 2 * (m$npar - 1))
print(rbind(full = sm(full, full$scale), reduced = sm(red, red$scale),
            inv_gauss = sm(ig, sqrt(ig$scale)) * c(1, 2, 1, 1, 1)), digits = 7)
```

```
##                scale   scale_t    logLik      AIC AIC_frees
## full      1.40878998  9.779048 -1558.667 3165.334  3163.334
## reduced   1.27969093  9.853830 -1567.928 3165.856  3163.856
## inv_gauss 0.02635282 17.720045 -1669.022 3368.044  3366.044
```

Every entry of FREES's Table 13.6 is reproduced. FREES's "Scale" is the shape $\hat{\nu}$ (1.409 and 1.280), and for the inverse Gaussian it is $\sqrt{\hat{\phi}} = 0.026$, whose t-value 17.72 is twice that of $\hat{\phi}$. FREES's AIC omits the scale parameter from the count: its 3163.34 is $-2(-1558.67) + 2 \cdot 23$ with the rounded log-likelihood, and counting the scale gives 3165.33. The inverse Gaussian column is a fit to all 157 observations, although the script fits that model with the largest expenditure removed, which would make its log-likelihood incomparable with the gamma's. The joint observed information is block-diagonal at the maximum, since $\partial^2 L / \partial \boldsymbol{\beta}\, \partial \nu = \sum_i (y_i/\hat{\mu}_i - 1)\mathbf{x}_i = \mathbf{0}$ by the score equations of Proposition 8.45(ii).

Example 8.16 (Observed, expected and robust standard errors)**.** For the full gamma model, the t-ratios of `COUNTIP` and `insure` from the observed information (FREES), the expected information with $\phi = 1/\hat{\nu}$, R's `summary` (expected information with the Pearson $\hat{\phi}$), and the sandwich:

```
X <- full$X; yy <- full$y; bF <- full$b; muF <- exp(drop(X %*% bF))
nu <- full$scale; Gobs <- nu * crossprod(X, (yy / muF) * X)
Gexp <- nu * crossprod(X); Fsw <- nu^2 * crossprod(X, (yy / muF - 1)^2 * X)
tt <- function(V) bF / sqrt(diag(V))
j <- c("COUNTIP", "insure")
```

```
round(rbind(observed = tt(solve(Gobs))[j], expected = tt(solve(Gexp))[j],
            glm_summary = coef(summary(gF))[j, 3],
            sandwich = tt(solve(Gexp, Fsw) %*% solve(Gexp))[j]), 3)
```

```
##              COUNTIP insure
## observed       6.155  3.068
## expected       7.495  3.181
## glm_summary    7.295  3.097
## sandwich       4.863  3.362
```

```
c(pearson_phi = summary(gF)$dispersion, one_over_nu = 1 / nu)
```

```
## pearson_phi one_over_nu
##      0.7492      0.7098
```

By Theorem 8.46 the third is consistent under the gamma model, and so are the first two provided $1/\hat{\nu}$ is a consistent estimate of ϕ, which is not proved here; the consistency of the sandwich when the variance function is wrong is likewise not proved here for gamma regression (Proposition 8.20 treats the Poisson case). Here they range from 4.86 to 7.49 for `COUNTIP`. Proposition 8.45(iii) weights observation i by $y_i/\hat{\mu}_i$ in the observed information, and a few very large expenditures dominate it.

8 Residuals

FREES §13.5. SRM learning outcome 2(d).

Definition 8.47 (Residuals; FREES §13.5)**.** For a generalised linear model with fitted means $\hat{\mu}_i$:

(i) the **raw** residual is $y_i - \hat{\mu}_i$;

(ii) the **Pearson** residual is $r_i^P = \sqrt{w_i}(y_i - \hat{\mu}_i)/\sqrt{v(\hat{\mu}_i)}$;

(iii) the **deviance** residual is $r_i^D = \operatorname{sign}(y_i - \hat{\mu}_i)\sqrt{d_i}$, with d_i of (16);

(iv) for a known transform h, $\{h(y_i) - \mathsf{E}\, h(y_i)\}/\sqrt{\operatorname{var} h(y_i)}$ evaluated at $\hat{\mu}_i$; h chosen to make $h(y_i)$ nearly normal gives the **Anscombe** residual, and h chosen to make $\operatorname{var} h(y_i)$ nearly constant a **variance-stabilising** one, such as $2(\sqrt{y} - \sqrt{\mu})$ for the Poisson.

A **Cox–Snell** residual is $R(y_i; \mathbf{x}_i, \hat{\boldsymbol{\theta}})$ for a function R making $R(y_i; \mathbf{x}_i, \boldsymbol{\theta})$ independent and identically distributed under the model; the Pearson residual at the true parameter is an example only up to its first two moments.

Proposition 8.48 (Properties; FREES §13.5)**.** At the true means, $\mathsf{E}\, r_i^P = 0$ and $\operatorname{var} r_i^P = \phi$. Always $\sum_i (r_i^P)^2 = \sum_i w_i(y_i - \hat{\mu}_i)^2 / v(\hat{\mu}_i)$, the Pearson statistic $\sum_i w_i(y_i - \hat{\mu}_i)^2/\{\phi v(\hat{\mu}_i)\}$ (remark after Theorem 8.41) times ϕ, and $\sum_i (r_i^D)^2 = D$.

Proof. $\operatorname{var} y_i = \phi v(\mu_i)/w_i$ (Definition 8.34); the sums are Definition 8.43 and the definitions above. □

Example 8.17 (Transforms of the Poisson, gamma and binomial; FREES Table 13.7)**.** Moments by summation over the Poisson and binomial mass functions, and by numerical integration for the gamma, for the square-root transform and FREES's Anscombe transform $1.5\lambda^{-1/6}\{y^{2/3} - (\lambda^{2/3} - \lambda^{-1/3}/9)\}$:

```
mom <- function(lam) { j <- 0:qpois(1 - 1e-15, lam); pj <- dpois(j, lam)
  v <- function(h) sum((h - sum(h * pj))^2 * pj)
  an <- 1.5 * lam^(-1/6) * (j^(2/3) - (lam^(2/3) - lam^(-1/3) / 9))
  c(lambda = lam, var_2sqrt = v(2 * sqrt(j)), mean_anscombe = sum(an * pj),
    var_anscombe = v(an)) }
round(t(sapply(c(0.5, 1, 5, 20, 100), mom)), 4)
```

```
##      lambda var_2sqrt mean_anscombe var_anscombe
## [1,]    0.5     1.240       -0.0599        1.004
## [2,]    1.0     1.609       -0.0779        1.184
## [3,]    5.0     1.144       -0.0081        1.055
## [4,]   20.0     1.020       -0.0007        1.009
## [5,]  100.0     1.004       -0.0001        1.002
```

```r
gam <- function(al) {
  a <- function(y) 3 * al^(1/6) * (y^(1/3) - (al^(1/3) - al^(-2/3) / 9))
  m <- integrate(function(y) a(y) * dgamma(y, al), 0, Inf, rel.tol = 1e-10)$value
  c(alpha = al, mean_anscombe = m,
    var_anscombe = integrate(function(y) (a(y) - m)^2 * dgamma(y, al), 0, Inf,
                             rel.tol = 1e-10)$value) }
round(t(sapply(c(1, 2, 5, 20, 100), gam)), 4)
```

```
##      alpha mean_anscombe var_anscombe
## [1,]     1        0.0123       0.9480
## [2,]     2        0.0024       0.9860
## [3,]     5        0.0002       0.9978
## [4,]    20        0.0000       0.9999
## [5,]   100        0.0000       1.0000
```

```r
bin <- function(m, p) { y <- 0:m; py <- dbinom(y, m, p)
  h <- 2 * sqrt(m) * (asin(sqrt(y / m)) - asin(sqrt(p))); mu <- sum(h * py)
  c(m = m, p = p, mean_arcsine = mu, var_arcsine = sum((h - mu)^2 * py)) }
round(t(mapply(bin, c(5, 20, 100, 5, 20, 100), c(0.5, 0.5, 0.5, 0.1, 0.1, 0.1))), 4)
```

```
##         m   p mean_arcsine var_arcsine
## [1,]    5 0.5       0.0000       1.545
## [2,]   20 0.5       0.0000       1.056
## [3,]  100 0.5       0.0000       1.010
## [4,]    5 0.1      -0.5011       1.353
## [5,]   20 0.1      -0.2544       1.546
## [6,]  100 0.1      -0.0703       1.043
```

Both variances approach 1 as λ grows (1.004 and 1.002 at $\lambda = 100$), and neither is close to 1 at $\lambda = 1$ (1.609 and 1.184). FREES's gamma row, $3\alpha^{1/6}\{(y/\gamma)^{1/3} - (\alpha^{1/3} - \alpha^{-2/3}/9)\}$, is evaluated with $\gamma = 1$ (the transform is free of the scale): its mean is within 0.013 of 0 and its variance within 0.052 of 1 already at $\alpha = 1$. For the binomial, the variance-stabilising arcsine transform $2\sqrt{m}\{\sin^{-1}\sqrt{y/m} - \sin^{-1}\sqrt{p}\}$ has variance near 1 only for large m, and more slowly when p is far from $\frac{1}{2}$ (1.04 at $m = 100$, $p = 0.1$). For the gamma model of Example 8.15, Figure 8.1 plots

the deviance residuals against normal quantiles.

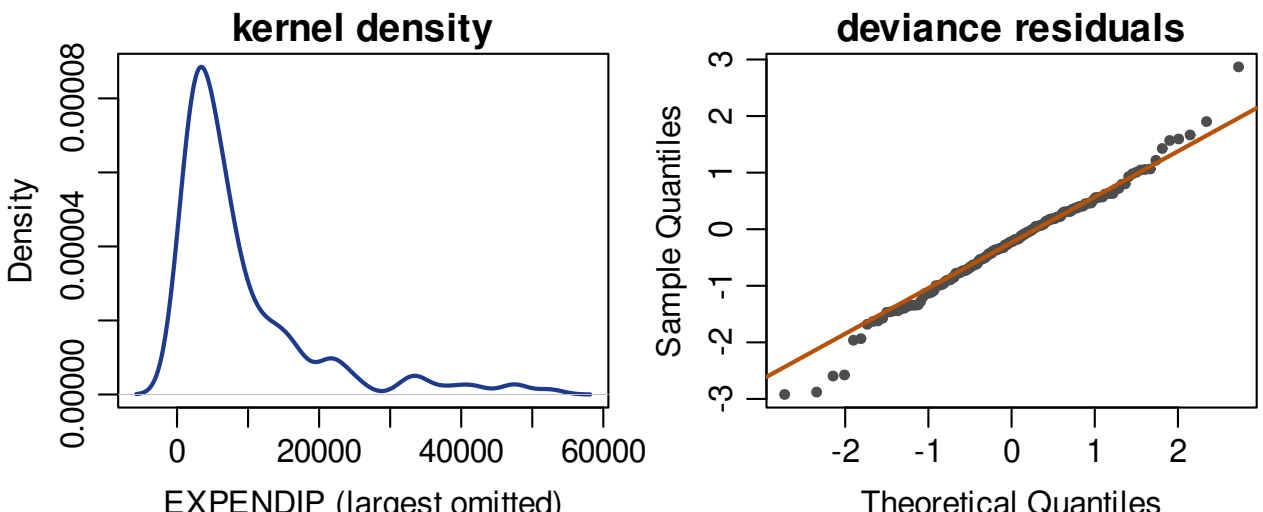


Figure 8.1: Left: kernel density of the positive inpatient expenditures without the largest (FREES Figure 13.1). Right: normal quantile plot of the deviance residuals of the reduced gamma model of Example 8.15.

9 The Tweedie Distribution

FREES §13.6. SRM learning outcome 2(b).

Lemma 8.49 (Sum of gammas)**.** If $X \sim \text{Gamma}(\alpha_1, \gamma)$ and $Y \sim \text{Gamma}(\alpha_2, \gamma)$ are independent, with density $\gamma^\alpha u^{\alpha-1} e^{-\gamma u}/\Gamma(\alpha)$ on $(0, \infty)$, then $X + Y \sim \text{Gamma}(\alpha_1 + \alpha_2, \gamma)$.

Proof. First, $\Gamma(\alpha_1)\Gamma(\alpha_2) = \Gamma(\alpha_1+\alpha_2)B(\alpha_1, \alpha_2)$ with $B(\alpha_1, \alpha_2) = \int_0^1 t^{\alpha_1-1}(1-t)^{\alpha_2-1}\mathrm{d}t$: writing the product as a double integral, substituting $u = st$, $v = s(1-t)$ with Jacobian s and using Fubini–Tonelli,

$$\int_0^\infty \int_0^\infty u^{\alpha_1-1} v^{\alpha_2-1} e^{-u-v} \mathrm{d}u\, \mathrm{d}v = \int_0^\infty s^{\alpha_1+\alpha_2-1} e^{-s} \mathrm{d}s \int_0^1 t^{\alpha_1-1}(1-t)^{\alpha_2-1} \mathrm{d}t.$$

For $s > 0$ the convolution density is $\gamma^{\alpha_1+\alpha_2} e^{-\gamma s} \{\Gamma(\alpha_1)\Gamma(\alpha_2)\}^{-1} \int_0^s u^{\alpha_1-1}(s-u)^{\alpha_2-1}\mathrm{d}u$, and the substitution $u = st$ makes the integral $s^{\alpha_1+\alpha_2-1} B(\alpha_1, \alpha_2)$. □

Theorem 8.50 (The Tweedie as a compound Poisson–gamma; FREES (13.9)–(13.12))**.** Let $N \sim \text{Pois}(\lambda)$ and $Y_1, Y_2, \ldots$ independent $\text{Gamma}(\alpha, \gamma)$, independent of N, and $T = \sum_{j \leqslant N} Y_j$, an empty sum being 0. For $\mu > 0$, $\phi > 0$ and $1 < p < 2$ put

$$\lambda = \frac{\mu^{2-p}}{\phi(2-p)}, \qquad \alpha = \frac{2-p}{p-1}, \qquad \frac{1}{\gamma} = \phi(p-1)\mu^{p-1}. \tag{17}$$

Then $\mathsf{P}(T=0) = e^{-\lambda}$, $\mathsf{E}\,T = \mu$, $\operatorname{var} T = \phi\mu^p$, and with respect to the measure $\delta_0 +$ Lebesgue measure on $(0,\infty)$ (the measure of Definition 8.32, there called λ), T has the density (12) with

$$\theta = \frac{\mu^{1-p}}{1-p}, \quad b(\theta) = \frac{\mu^{2-p}}{2-p}, \quad S(y,\phi) = \ln \sum_{n=1}^{\infty} \frac{1}{n!\,\Gamma(n\alpha)\,y} \left\{ \frac{y^\alpha}{(p-1)^\alpha \phi^{1+\alpha}(2-p)} \right\}^n \ (y>0), \tag{18}$$

and $S(0,\phi) = 0$. So the Tweedie is a linear exponential family with $v(\mu) = \mu^p$.

Proof. Every $Y_j > 0$, so $T = 0$ exactly when $N = 0$. By the prerequisites' corollary **random sums**, with $\mathsf{E}\,Y_1 = \alpha/\gamma$ and $\operatorname{var} Y_1 = \alpha/\gamma^2$, $\mathsf{E}\,T = \lambda\alpha/\gamma$ and $\operatorname{var} T = \lambda\alpha(1+\alpha)/\gamma^2$. Now

$\alpha/\gamma = \phi(2-p)\mu^{p-1}$, so $\mathsf{E}\,T = \mu$, and $1+\alpha = 1/(p-1)$, so $\operatorname{var} T = \mu\cdot\phi(p-1)\mu^{p-1}/(p-1) = \phi\mu^p$. Given $N = n \geqslant 1$, $Y_1 + \cdots + Y_n \sim \text{Gamma}(n\alpha, \gamma)$ by Lemma 8.49 and induction, so for Borel $A \subset (0,\infty)$, by Fubini–Tonelli,

$$\mathsf{P}(T \in A) = \int_A \sum_{n\geqslant 1} e^{-\lambda}\frac{\lambda^n}{n!}\frac{\gamma^{n\alpha}}{\Gamma(n\alpha)} y^{n\alpha-1}e^{-\gamma y}\,\mathrm{d}y,$$

which is FREES (13.9). With (17), $\lambda+\gamma y = \{b(\theta)-y\theta\}/\phi$, since $\gamma y = y\mu^{1-p}/\{\phi(p-1)\} = -y\theta/\phi$ and $\lambda = b(\theta)/\phi$; and $\lambda\gamma^\alpha = \mu^{(2-p)-\alpha(p-1)}/\{\phi(2-p)(p-1)^\alpha\phi^\alpha\}$, where the exponent of μ is $(2-p)-(2-p) = 0$. Hence the density is $\exp\{(y\theta - b(\theta))/\phi + S(y,\phi)\}$ with S as in (18), and at $y = 0$ the mass $e^{-\lambda} = e^{-b(\theta)/\phi}$ is the same expression with $S(0,\phi) = 0$. As $\mu = \{(1-p)\theta\}^{1/(1-p)}$, $\mathrm{d}\theta/\mathrm{d}\mu = \mu^{-p}$, so $b'(\theta) = \mu^{1-p}\mu^p = \mu$ and $b''(\theta) = \mu^p$. □

Remark (FREES §13.6, exercise 13.3)**.** $S(y,\phi)$ depends on p through α but not on μ, as FREES's exercise 13.3 requires. For $1 < p < 2$ the law has an atom at 0 and a density on $(0,\infty)$ (Figure 8.2): a policy's aggregate loss, zero without a claim. The variance function μ^p is the Poisson's at $p = 1$ and the gamma's at $p = 2$ (FREES Table 13.1), and $1 < p < 2$ lies between them.

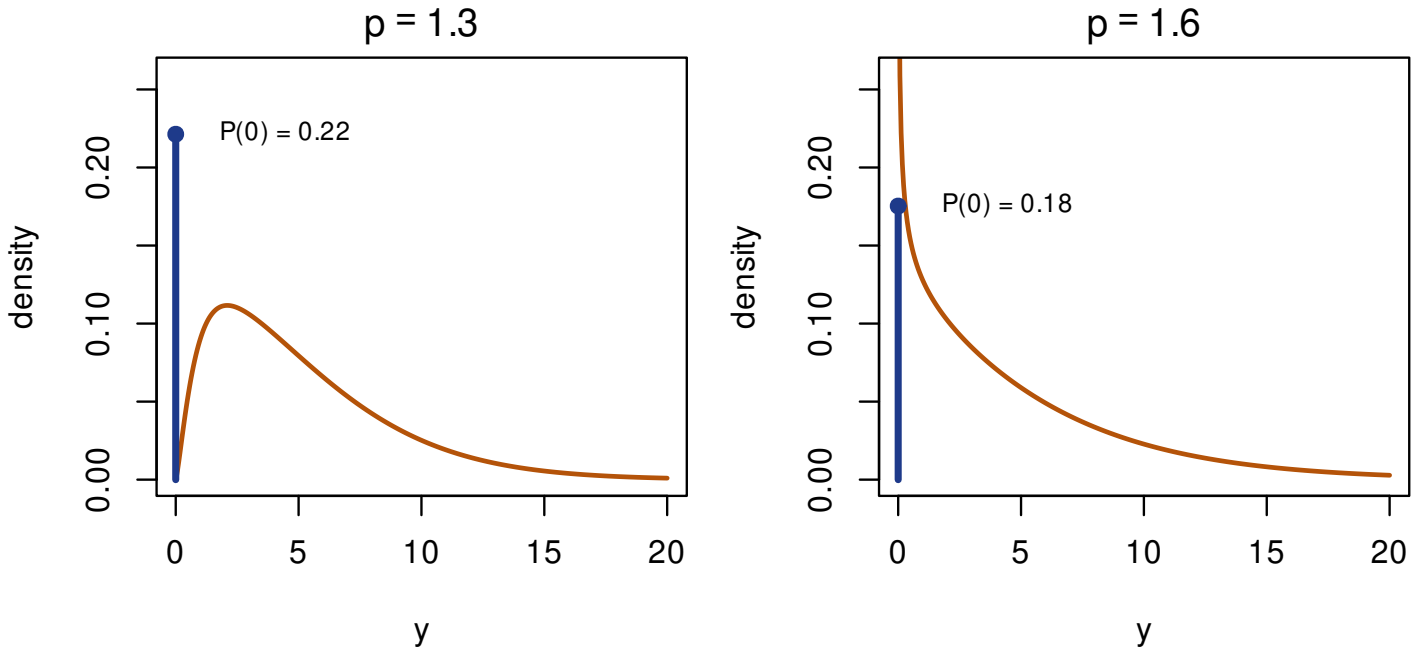


Figure 8.2: Tweedie laws with $\mu = 4$, $\phi = 2.5$. The spike is the atom $e^{-\lambda}$ at 0 of Theorem 8.50, the curve the density of the positive part.

10 Computation in R

FREES chapters 12–13 scripts. SRM learning outcomes 2(d), 2(f). Beyond the syllabus: ISLR §4.7.7, the chapter's lab, which neither reading list includes.

10.1 The ISLR lab (beyond the syllabus)

```
mod.lm <- lm(bikers ~ mnth + hr + workingday + temp + weathersit, data = Bikeshare)
c(feb_treatment = coef(mod.lm)[["mnthFeb"]], jan_sum_coding = coef(mlm)[["mnth1"]],
  sum_sq_prediction_diff = sum((predict(mod.lm) - predict(mlm))^2))
```

```
##          feb_treatment          jan_sum_coding
##              6.845e+00              -4.609e+01
## sum_sq_prediction_diff
##              2.090e-21
```

```
all.equal(predict(mod.lm), predict(mlm))
```

```
## [1] TRUE
```

```
pl <- predict(mlm); pq <- predict(mpo, type = "response")
dec <- cut(pl, quantile(pl, 0:10 / 10), include.lowest = TRUE)
```

```
round(tapply(pq - pl, dec, mean), 1)
```

```
## [-144,1.93] (1.93,49.1] (49.1,74.7]  (74.7,106]   (106,142]
##        51.4         0.0       -24.3       -19.8       -17.0
##   (142,176]   (176,208]   (208,237]   (237,288]   (288,441]
##       -14.6        -7.3        -0.5        -0.1        32.2
```

Under treatment coding the February coefficient 6.845 is relative to January; under `contr.sum` the January coefficient −46.087 is relative to the average month, the twelve coefficients summing to zero (Figure 8.3). The fitted values are the same, as both codings span the same column space. By decile of the least squares prediction, the Poisson prediction exceeds it by 51.4 in the lowest decile and 32.2 in the highest, is within 0.1 of it in the second and ninth, and falls below it in the third to eighth: ISLR's "larger for very low or very high levels of ridership".

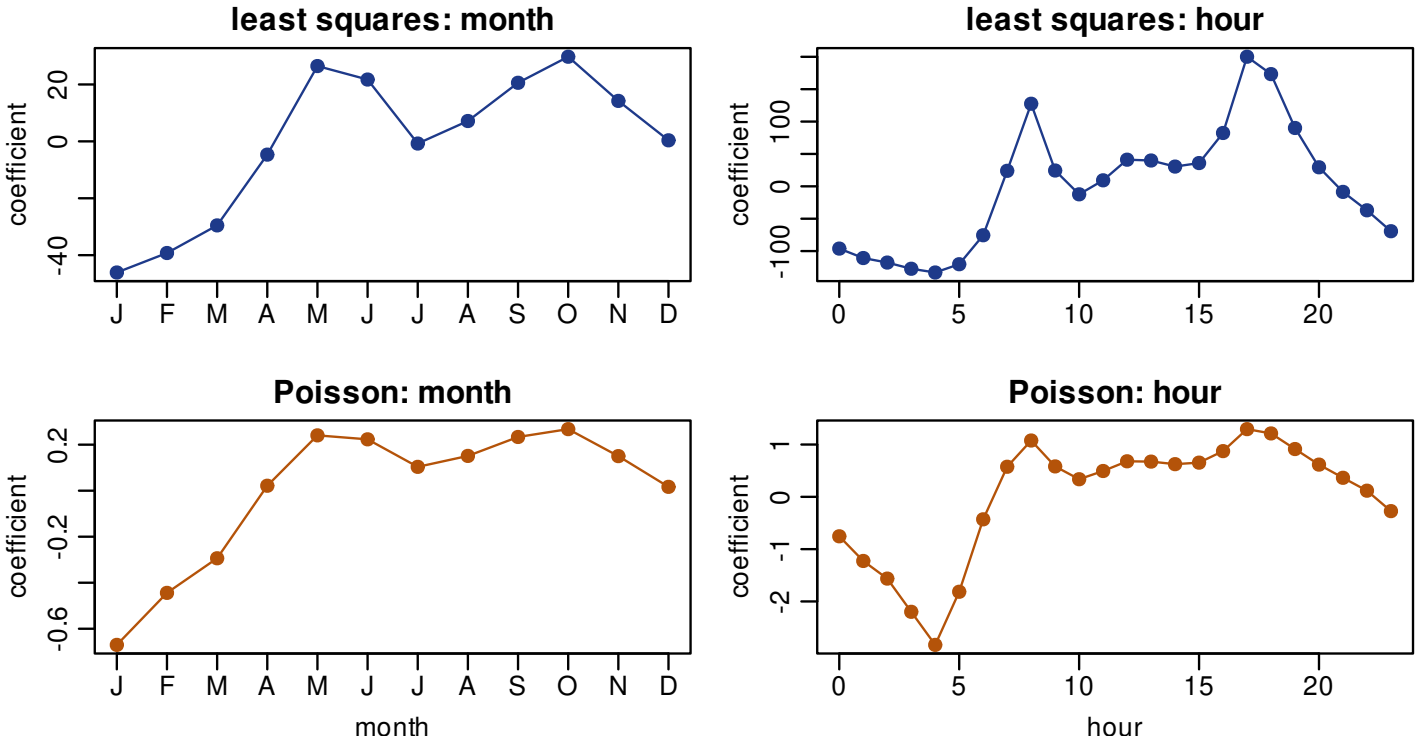


Figure 8.3: Month and hour coefficients under `contr.sum` coding, the last level being minus the sum of the others (ISLR Figures 4.13 and 4.15). Top: least squares, in bikers per hour. Bottom: Poisson, on the log scale.

10.2 Formulas against package output

Example 8.18 (Poisson score, information and IRLS; Theorems 8.11, 8.40)**.** With `glm`'s convergence tightened (Unit 7):

```
ctl <- glm.control(epsilon = 1e-14, maxit = 100)
m3t <- glm(fS, offset = LNWEIGHT, family = poisson, data = S, control = ctl)
X3 <- model.matrix(m3t)[, !is.na(coef(m3t))]; b3t <- coef(m3t)[!is.na(coef(m3t))]
mu3t <- fitted(m3t); off <- S$LNWEIGHT
irls <- function(X, y, off, link_inv, dmu, vfun, eta0) {
  eta <- eta0
  repeat { mu <- link_inv(eta); d <- dmu(eta); W <- d^2 / vfun(mu)
    z <- eta - off + (y - mu) / d
    b <- drop(solve(crossprod(X, W * X), crossprod(X, W * z)))
    eta_new <- drop(X %*% b) + off
    if (max(abs(eta_new - eta)) < 1e-13) return(b)
    eta <- eta_new } }
b_irls <- irls(X3, y, off, exp, exp, identity, log(y + 0.5))
signif(c(
  max_abs_score = max(abs(crossprod(X3, y - mu3t))),
```

```r
  se_manual_minus_glm = max(abs(sqrt(diag(solve(crossprod(X3, mu3t * X3)))) -
                                coef(summary(m3t))[, 2])),
  irls_minus_glm = max(abs(b_irls - b3t)), 3)
```

```
##       max_abs_score se_manual_minus_glm      irls_minus_glm
##            1.34e-13            1.09e-09           8.13e-14
```

Example 8.19 (A non-canonical link: gamma with logarithmic link)**.** IRLS of Theorem 8.40 with $v(\mu) = \mu^2$ and $\mathrm{d}\mu/\mathrm{d}\eta = \mu$, so $W_i = 1$ and $z_i = \eta_i + (y_i - \mu_i)/\mu_i$, against **glm**, and the score of Proposition 8.45(ii) at **glm**'s estimate:

```r
gRt <- glm(fR, data = D, family = Gamma(link = "log"), control = ctl)
XR <- model.matrix(gRt)
bR_irls <- irls(XR, D$EXPENDIP, 0, exp, exp, function(m) m^2, log(D$EXPENDIP))
signif(c(irls_minus_glm = max(abs(bR_irls - coef(gRt))),
         score_at_glm = max(abs(crossprod(XR, D$EXPENDIP / fitted(gRt) - 1)))), 3)
```

```
## irls_minus_glm   score_at_glm
##    0.000000104    0.000055400
```

Example 8.20 (Deviances and residuals; Propositions 8.44, 8.48)**.**

```
dP <- 2 * (ifelse(y == 0, 0, y * log(y / mu3t)) - (y - mu3t))
yg <- D$EXPENDIP; mg <- fitted(gRt); dG <- 2 * (-log(yg / mg) + (yg - mg) / mg)
signif(c(poisson_D_minus_glm = sum(dP) - deviance(m3t),
         gamma_D_minus_glm = sum(dG) - deviance(gRt),
         deviance_resid = max(abs(sign(y - mu3t) * sqrt(pmax(dP, 0)) -
                                  residuals(m3t, "deviance"))),
         pearson_resid = max(abs((y - mu3t) / sqrt(mu3t) - residuals(m3t,
          ↪  "pearson")))), 3)
```

```
## poisson_D_minus_glm   gamma_D_minus_glm      deviance_resid
##                   0                   0                   0
##       pearson_resid
##                   0
```

Example 8.21 (The deviance of a correct Poisson model; FREES §13.3.3)**.** Intercept-only Poisson samples of size $n = 2000$, 200 replications for each mean:

```
set.seed(21)
t(sapply(c(0.3, 1, 5, 50), function(m) {
```

```
  Dv <- replicate(200, deviance(glm(rpois(2000, m) ~ 1, family = poisson)))
  Xv <- replicate(200, { yy <- rpois(2000, m); sum((yy - mean(yy))^2) / mean(yy) })
  round(c(mu = m, mean_deviance = mean(Dv), mean_pearson = mean(Xv),
          n_minus_1 = 1999), 1)
}))
```

```
##        mu mean_deviance mean_pearson n_minus_1
## [1,]  0.3          1683         1991      1999
## [2,]  1.0          2296         2001      1999
## [3,]  5.0          2093         1996      1999
## [4,] 50.0          2009         2009      1999
```

The mean deviance is 1683 at $\mu = 0.3$ and 2296 at $\mu = 1$, against 1999; the Pearson statistic stays within 10 of 1999 at every mean (Proposition 8.17).

11 Exercises

Exercise 8.1 (FREES exercise 12.1)**.** Show that the log-likelihood (2) has a maximum at $\hat{\mu} = \bar{y}$.

Solution. Proposition 8.3: when $\bar{y} > 0$, $L'(\mu) = n(\bar{y}/\mu - 1)$ is positive below $\bar{y}$ and negative above it, and $L'' = -n\bar{y}/\mu^2 < 0$, so $\bar{y}$ is the unique maximiser. When every $y_i = 0$, $L(\mu) = -n\mu$

has supremum 0 at the excluded $\mu = 0$ and no maximiser in $(0, \infty)$. ■

Exercise 8.2 (FREES exercise 12.2)**.** For the data of FREES Table 12.1, confirm that the Pearson statistic is 41.98.

Solution. Example 8.2 computes 41.984 from FREES's five cells $\{0\}, \dots, \{4\}$ and the fitted counts $n\, e^{-\bar{y}} \bar{y}^j / j!$: the cells contribute $(6996 - 6977.86)^2/6977.86 = 0.05$, $(455 - 487.69)^2/487.69 = 2.19$, $(28 - 17.043)^2/17.043 = 7.04$, $(4 - 0.3971)^2/0.3971 = 32.69$ and 0.0069. Cell $\{3\}$ supplies 78% of the total. ■

Exercise 8.3 (FREES exercise 12.3)**.** In a Poisson regression with an intercept, let $e_i = y_i - \hat{\mu}_i$. Show that (a) the average residual is 0; (b) the correlation between the residuals and each explanatory variable is 0.

Solution. (a) Corollary 8.12 with $x_{i0} = 1$: $\sum_i e_i = 0$.
(b) For each j, $\sum_i x_{ij} e_i = 0$ by the same corollary, so the sample covariance $n^{-1} \sum_i (x_{ij} - \bar{x}_j)(e_i - \bar{e}) = n^{-1} \sum_i x_{ij} e_i - \bar{x}_j \bar{e} = 0$ by (a). The correlation is this covariance divided by the two standard deviations, hence 0 whenever both are positive, and undefined otherwise. Example 8.18 prints

$\max_j |\sum_i x_{ij} e_i|$ at the level of rounding error. ■

Exercise 8.4 (FREES exercise 12.4)**.** (a) Let $y_1, \dots, y_n$ be independent $\mathrm{NB}(r, p)$. Determine the maximum likelihood estimators. (b) Do the same in the parametrisation $\sigma = 1/r$, $\mu = r(1-p)/p$. (c) For independent $y_i \sim \mathrm{NB}(\mu_i, \sigma)$ with $\mu_i = \exp(\mathbf{x}_i'\boldsymbol{\beta})$, determine the score in σ and $\boldsymbol{\beta}$.

Solution. (a) Assume $\bar{y} > 0$; if every $y_i = 0$, the p-derivative nr/p is positive on $(0,1)$ and no maximiser exists. With $N_r(y) = \ln\Gamma(y+r) - \ln\Gamma(r) - \ln y!$, the log-likelihood is $\sum_i N_r(y_i) + nr\ln p + n\bar{y}\ln(1-p)$. For fixed r its p-derivative $nr/p - n\bar{y}/(1-p)$ is strictly decreasing and vanishes at $\hat{p}(r) = r/(r+\bar{y})$, the maximiser in p. Substituting, the profile log-likelihood is

$$\ell_p(r) = \sum_i N_r(y_i) + nr\ln\frac{r}{r+\bar{y}} + n\bar{y}\ln\frac{\bar{y}}{r+\bar{y}},$$

and since $\Gamma(y+r)/\Gamma(r) = \prod_{l=0}^{y-1}(r+l)$,

$$\ell_p'(r) = \sum_i \sum_{l=0}^{y_i-1} \frac{1}{r+l} + n\ln\frac{r}{r+\bar{y}}.$$

$\hat r$ is a root of ℓ_p', found numerically (the root with the largest ℓ_p, provided that value exceeds the Poisson log-likelihood at $\bar y$; otherwise none exists), and $\hat p = \hat r/(\hat r+\bar y)$. If some $y_i \geqslant 1$, $\ell_p(r) \to -\infty$ as $r \downarrow 0$, because $\ln \prod_l (r+l)$ contains $\ln r$ and the other terms stay bounded. So if ℓ_p' has no root it is positive throughout, ℓ_p increases, and its supremum is the limit $r \to \infty$, the Poisson log-likelihood at $\bar y$ (Corollary 8.23); no maximiser exists.

(b) The map $(r,p) \mapsto (\sigma,\mu) = (1/r, r(1-p)/p)$ is a bijection of $(0,\infty)\times(0,1)$ onto $(0,\infty)^2$, so the maximiser maps to the maximiser: $\hat\sigma = 1/\hat r$ and $\hat\mu = \hat r(1-\hat p)/\hat p = \hat r \cdot \bar y/\hat r = \bar y$.

(c) With $r = 1/\sigma$, $\ln f(y) = \ln\Gamma(y+1/\sigma) - \ln\Gamma(1/\sigma) - \ln y! - \sigma^{-1}\ln(1+\sigma\mu) + y\ln(\sigma\mu) - y\ln(1+\sigma\mu)$. Then $\partial \ln f/\partial\mu = (y-\mu)/\{\mu(1+\sigma\mu)\}$, and with $\partial\mu_i/\partial\boldsymbol\beta = \mu_i \mathbf{x}_i$,

$$\frac{\partial L}{\partial \boldsymbol\beta} = \sum_i \frac{y_i-\mu_i}{1+\sigma\mu_i}\mathbf{x}_i, \qquad \frac{\partial L}{\partial \sigma} = \sum_i \Big[\frac{1}{\sigma^2}\Big\{\ln(1+\sigma\mu_i) - \sum_{l=0}^{y_i-1}\frac{1}{1/\sigma+l}\Big\} + \frac{y_i-\mu_i}{\sigma(1+\sigma\mu_i)}\Big],$$

the last term collecting $-\mu_i/\{\sigma(1+\sigma\mu_i)\} + y_i/\sigma - y_i\mu_i/(1+\sigma\mu_i)$. Both vanish at the `glm.nb` fit of Example 8.10, and (a) on the Singapore counts without covariates:

```
mn <- fitted(nb3); sg <- 1 / nb3$theta; Xn <- model.matrix(nb3)
inner <- sapply(y, function(k) if (k == 0) 0 else sum(1 / (1 / sg + 0:(k - 1))))
```

```r
dsig <- sum((log(1 + sg * mn) - inner) / sg^2 + (y - mn) / (sg * (1 + sg * mn)))
dprof <- function(r) sum(sapply(y, function(k) if (k == 0) 0 else sum(1 / (r + 0:(k - 1))))) +
  n * log(r / (r + ybar))
r_hat <- uniroot(dprof, c(0.01, 100))$root
signif(c(score_beta = max(abs(crossprod(Xn, (y - mn) / (1 + sg * mn)))),
         score_sigma = dsig, r_hat = r_hat, p_hat = r_hat / (r_hat + ybar),
         glm_nb_r = glm.nb(y ~ 1)$theta), 4)
```

```
##  score_beta score_sigma       r_hat       p_hat    glm_nb_r
##   1.618e-09  -9.716e-09   8.740e-01   9.260e-01   8.740e-01
```

Exercise 8.5 (FREES exercise 12.5)**.** For the MEPS data, with $y =$ `COUNTOP`, the number of outpatient visits, and the explanatory variables of FREES §11.4: (a) tabulate and summarise y and compare its mean and variance; (b) tabulate its means by the levels of each explanatory variable; (c) fit a Poisson model without covariates and compute the individual-level Pearson statistic; (d) fit the Poisson model with the covariates: (i) significance, (ii) the `GENDER` coefficient, (iii) the Pearson statistic and a choice between the models, (iv) the quasi-likelihood refit; (e) fit

the negative binomial: (i) significance, (ii) the Pearson statistic and AIC, (iii) a likelihood ratio test for income; (f) fit a logistic regression for $\{y > 0\}$ and compare it with (e).

Solution. FREES's parts (d)(iii) and (e)(ii) refer to the Pearson statistic of "part (b)"; it is part (c).

```
H$RACE[H$RACE == "OTHER"] <- "WHITE"; H$RACE <- relevel(factor(H$RACE), ref =
 ↪  "WHITE")
H$REGION <- relevel(factor(H$REGION), ref = "WEST")
H$EDUC <- relevel(factor(H$EDUC), ref = "LHIGHSC")
H$PHSTAT <- relevel(factor(H$PHSTAT), ref = "EXCE")
H$INCOME <- relevel(factor(H$INCOME), ref = "HINCOME")
yo <- H$COUNTOP; table(cut(yo, c(-1, 0, 1, 2, 5, 10, 20, 50, 200)))
```

```
##
##   (-1,0]     (0,1]     (1,2]     (2,5]    (5,10]   (10,20]
##      648       284       190       343       251       171
##  (20,50] (50,200]
##       86        27
```

```
c(mean = mean(yo), variance = var(yo), median = median(yo), max = max(yo))
```

```
##     mean variance   median      max
##    5.675  156.251    2.000  167.000
```

```
sapply(c("GENDER", "RACE", "REGION", "EDUC", "PHSTAT", "MNHPOOR", "ANYLIMIT",
         "INCOME", "insure"), function(v) round(tapply(yo, H[[v]], mean), 2))
```

```
## $GENDER
##    0    1
## 4.25 6.95
##
## $RACE
## WHITE ASIAN BLACK NATIV
##  5.64  2.98  6.36  9.91
##
## $REGION
##      WEST   MIDWEST NORTHEAST     SOUTH
##      4.85      7.09      5.70      5.54
##
## $EDUC
## LHIGHSC COLLEGE HIGHSCH
##    4.84    6.73    5.58
```

```
##
## $PHSTAT
##  EXCE  FAIR  GOOD  POOR  VGOO
##  3.19  9.45  6.35 17.40  4.44
##
## $MNHPOOR
##     0     1
##  5.07 13.21
##
## $ANYLIMIT
##     0     1
##  3.57 13.01
##
## $INCOME
## HINCOME LINCOME MINCOME   NPOOR    POOR
##    6.08    4.92    5.29    6.60    5.98
##
## $insure
##    0    1
## 2.84 6.49
```

(a) 648 of the 2000 persons have no visit, the median is 2 and the maximum 167: a long right tail. The variance 156.3 is 27.5 times the mean 5.67, far from equidispersion. (b) The means differ sharply by physical health (3.19 excellent, 17.40 poor), activity limitation (3.57 against 13.01), mental health, insurance (2.84 against 6.49) and sex (4.25 men, 6.95 women), and less by region and income.

```
f0 <- COUNTOP ~ GENDER + RACE + REGION + EDUC + PHSTAT + MNHPOOR + ANYLIMIT + INCOME
 ↪  + insure
pearson <- function(m, v) sum((yo - fitted(m))^2 / v(fitted(m)))
p0 <- glm(COUNTOP ~ 1, family = poisson, data = H); p1 <- glm(f0, family = poisson,
 ↪  data = H)
q1 <- glm(f0, family = quasipoisson, data = H); n1 <- glm.nb(f0, data = H)
n2 <- glm.nb(update(f0, . ~ . - INCOME), data = H)
n2r <- glm(update(f0, . ~ . - INCOME), family = negative.binomial(n1$theta), data =
 ↪  H)
l1 <- glm(update(f0, I(COUNTOP > 0) ~ .), family = binomial, data = H)
round(cbind(pois_b = coef(p1), pois_z = coef(summary(p1))[, 3],
            quasi_t = coef(summary(q1))[, 3], nb_b = coef(n1),
            nb_z = coef(summary(n1))[, 3], logit_z = coef(summary(l1))[, 3]), 2)
```

```
##                 pois_b pois_z quasi_t  nb_b  nb_z logit_z
## (Intercept)       0.03   0.72    0.18 -0.22 -1.58   -6.69
## GENDER            0.42  20.94    5.17  0.61  9.16    9.03
## RACEASIAN        -0.52  -8.17   -2.02 -0.60 -3.53   -0.77
## RACEBLACK         0.05   1.96    0.48 -0.17 -1.79   -0.28
## RACENATIV         0.50   7.12    1.76  0.22  0.70    1.18
## REGIONMIDWEST     0.23   8.39    2.07  0.20  2.05    2.83
## REGIONNORTHEAST   0.03   0.90    0.22  0.00  0.01    0.86
## REGIONSOUTH      -0.05  -2.15   -0.53 -0.07 -0.84    1.47
## EDUCCOLLEGE       0.52  18.37    4.54  0.46  4.65    3.73
## EDUCHIGHSCH       0.21   8.37    2.07  0.24  2.85    1.78
## PHSTATFAIR        0.79  21.31    5.27  0.85  6.53    4.19
## PHSTATGOOD        0.57  18.58    4.59  0.66  7.27    4.79
## PHSTATPOOR        0.98  22.70    5.61  1.15  5.96    4.20
## PHSTATVGOO        0.27   8.69    2.15  0.43  4.80    4.61
## MNHPOOR           0.31  10.58    2.61  0.41  3.06    0.61
## ANYLIMIT          1.03  48.72   12.04  1.12 13.73    7.02
## INCOMELINCOME    -0.23  -6.96   -1.72 -0.47 -4.24   -3.33
## INCOMEMINCOME    -0.16  -6.31   -1.56 -0.15 -1.77   -3.24
## INCOMENPOOR       0.01   0.18    0.05 -0.15 -1.01   -1.85
```

```
## INCOMEPOOR       -0.21  -6.42   -1.59 -0.10 -0.91   -2.54
## insure            0.54  17.15    4.24  0.66  7.43    9.14
```

```
LRi <- 2 * (as.numeric(logLik(n1)) - as.numeric(logLik(n2)))
LRr <- 2 * (as.numeric(logLik(n1)) - as.numeric(logLik(n2r)))
round(c(X2_null = pearson(p0, identity), X2_poisson = pearson(p1, identity),
        X2_negbin = pearson(n1, function(m) m + m^2 / n1$theta),
        phi_quasi = summary(q1)$dispersion, AIC_null = AIC(p0),
        AIC_poisson = AIC(p1), AIC_negbin = AIC(n1), r_negbin = n1$theta,
        exp_gender = exp(coef(p1)[["GENDER"]]), LR_income = LRi,
        p_income = pchisq(LRi, 4, lower.tail = FALSE), LR_income_fixed_r = LRr,
        p_income_fixed_r = pchisq(LRr, 4, lower.tail = FALSE)), 3)
```

```
##          X2_null       X2_poisson         X2_negbin
##        55043.634        32397.773         3678.348
##        phi_quasi          AIC_null       AIC_poisson
##           16.372        29853.138         22645.963
##       AIC_negbin          r_negbin        exp_gender
##         9992.418             0.562             1.522
##        LR_income         p_income LR_income_fixed_r
##           18.963             0.001            19.037
```

```
##  p_income_fixed_r
##             0.001
```

(c) The Pearson statistic of the null Poisson model is 55,044 on 1999 degrees of freedom. (d)(i) Every coefficient but the intercept, `BLACK` ($z = 1.96$ to two decimals, 1.959), `NORTHEAST` and `NPOOR` has $|z| > 1.96$, twelve of them $|z| > 8$. (ii) Women have $e^{0.420} = 1.52$ times the expected visits of comparable men (Proposition 8.10(ii)). (iii) The Pearson statistic falls to 32,398 on 1979 degrees of freedom, $\hat{\phi} = 16.4$; the covariates help, but by Proposition 8.17 the Poisson variance is wrong by a factor of about 16, and the z values of (i) are not to be trusted. (iv) Under Theorem 8.19 every t is the z divided by $\sqrt{16.37} = 4.05$: `SOUTH`, `LINCOME`, `MINCOME`, income `POOR` and `NATIV` are no longer significant at 5% (`BLACK`, with $|z| < 1.96$, was not significant in (i)), while `GENDER`, health, `ANYLIMIT`, education, `insure`, `ASIAN`, `MIDWEST` and `MNHPOOR` remain so. (e)(i) With $\hat{r} = 0.562$ the significant variables are `GENDER`, `ASIAN`, `MIDWEST`, education, all health levels, `MNHPOOR`, `ANYLIMIT`, `LINCOME` and `insure`. (ii) The Pearson statistic with $v(\mu) = \mu + \mu^2/\hat{r}$ is 3678; the AIC falls from 29,853 (null) and 22,646 (Poisson) to 9992, so the negative binomial is preferred. (iii) Dropping the four income indicators gives LRT = 18.96, $p = 0.0008$ against χ^2_4, with r re-estimated in both fits by `glm.nb`; holding r at $\hat{r}$ in both fits gives LRT = 19.04, again

$p = 0.0008$. The prerequisites' theorem **likelihood ratio** covers the second version for a known r, under $|\mathbf{x}_i| \leqslant M$ and $n^{-1}\sum_i \mu_i \mathbf{x}_i\mathbf{x}_i'/(1+\sigma\mu_i) \to \mathbf{F}$ positive definite. Its (L1) is Proposition 8.25. The score $(y_i - \mu_i)\mathbf{x}_i/(1+\sigma\mu_i)$ has mean $\mathbf{0}$, variance $\mu_i\mathbf{x}_i\mathbf{x}_i'/(1+\sigma\mu_i)$ by (9), and bounded fourth moments since the μ_i are bounded, so the Lyapunov corollary gives (L2). The Hessian $\sum_i\{(y_i + r)r\mu_i/(r+\mu_i)^2\}\mathbf{x}_i\mathbf{x}_i'$ has mean $\sum_i \mu_i\mathbf{x}_i\mathbf{x}_i'/(1+\sigma\mu_i)$, so $\mathbf{G} = \mathbf{F}$, and converges after division by n by Chebyshev's inequality; since $re^\eta/(r+e^\eta)^2 \leqslant \frac{1}{4}$, the derivative of $(y_i+r)re^\eta/(r+e^\eta)^2$ in η is at most $(y_i+r)/4$ in absolute value, so the curvature condition holds with $L_n = M^3 n^{-1}\sum_i(y_i+r)/4$, bounded in probability by Markov's inequality as in Theorem 8.46; this is (L3). The restricted model satisfies the same conditions. Neither the re-estimation of r nor the use of the estimate $\hat{r}$ in place of a known r is covered here, so both p-values are approximations; income is significant by either.

(f) Every slope of the logistic regression of $\{y > 0\}$ has the sign of the negative binomial's except `SOUTH` ($z = 1.47$ against -0.84), insignificant in both. Significance differs: `ASIAN`, `MNHPOOR` and `HIGHSCH` matter for the number of visits and not for whether any occurs, `MINCOME` and income `POOR` the reverse. ■

Exercise 8.6 (FREES exercise 12.6)**.** Let $x_i \in \{0,1\}$, with n_0 observations at $x=0$ of mean $\bar y_0$ and $n_1 = n - n_0$ at $x=1$ of mean $\bar y_1$, and fit $\ln \mu_i = \beta_0 + \beta_1 x_i$. (i) Find the maximum likelihood estimators. (ii) Evaluate them for $n_0 = 10$, $n_1 = 90$, $\bar y_0 = 0.20$, $\bar y_1 = 0.05$. (iii) Find the information matrix.

Solution. (i) By Corollary 8.12 the score equations are $\sum_i (y_i - \mu_i) = 0$ and $\sum_i x_i (y_i - \mu_i) = 0$, that is $n_1 e^{\beta_0+\beta_1} = n_1 \bar y_1$ and $n_0 e^{\beta_0} = n_0 \bar y_0$. When $\bar y_0, \bar y_1 > 0$, $b_0 = \ln \bar y_0$ and $b_1 = \ln(\bar y_1 / \bar y_0)$, the unique maximiser by the strict concavity of Theorem 8.11 and the prerequisites' proposition **first-order condition**. If $\bar y_1 = 0$ no maximiser exists by Corollary 8.14; if $\bar y_0 = 0$, by Theorem 8.13(i) with $\mathbf{c} = (-1, 1)'$, for which $\mathbf{x}_i'\mathbf{c} = -1$ at $x_i = 0$ and 0 at $x_i = 1$. (ii) $b_0 = \ln 0.20 = -1.609$ and $b_1 = \ln 0.25 = -1.386$. (iii) By Theorem 8.11, with $\mu_0 = e^{\beta_0}$ and $\mu_1 = e^{\beta_0+\beta_1}$,

$$\mathcal{I}(\boldsymbol\beta) = \begin{pmatrix} n_0\mu_0 + n_1\mu_1 & n_1\mu_1 \\ n_1\mu_1 & n_1\mu_1 \end{pmatrix}, \qquad \mathcal{I}(\mathbf{b}) = \begin{pmatrix} 6.5 & 4.5 \\ 4.5 & 4.5 \end{pmatrix}$$

for the data of (ii), since $n_0\bar y_0 = 2$ and $n_1 \bar y_1 = 4.5$. ■

Exercise 8.7 (FREES exercise 13.1)**.** Verify the gamma entry of FREES Table 13.8: show that the gamma is a linear exponential family, give $(\theta, \phi, b(\theta), S(y, \phi))$ in terms of (α, γ), and check the mean and variance.

Solution. Proposition 8.35: $\theta = -\gamma/\alpha$, $\phi = 1/\alpha$, $b(\theta) = -\ln(-\theta)$ on $\theta < 0$, and $S(y, \phi) = -\phi^{-1} \ln \phi - \ln \Gamma(\phi^{-1}) + (\phi^{-1} - 1) \ln y$. By Theorem 8.33, $\mathsf{E}\, y = b'(\theta) = -1/\theta = \alpha/\gamma$ and $\operatorname{var} y = \phi b''(\theta) = \phi/\theta^2 = \alpha/\gamma^2$, the gamma's moments ($\mathsf{E}\, y = \int_0^\infty \gamma^\alpha y^\alpha e^{-\gamma y} \mathrm{d}y/\Gamma(\alpha) = \Gamma(\alpha + 1)/\{\gamma\Gamma(\alpha)\} = \alpha/\gamma$, and likewise $\mathsf{E}\, y^2 = \alpha(\alpha + 1)/\gamma^2$). FREES's table has $+\phi^{-1} \ln \phi$:

```
al <- 2.5; ga <- 0.7; yv <- c(0.5, 2, 6); th <- -ga / al; ph <- 1 / al
lef <- function(sgn) exp((yv * th + log(-th)) / ph + sgn * log(ph) / ph -
                         lgamma(1 / ph) + (1 / ph - 1) * log(yv))
signif(rbind(dgamma = dgamma(yv, al, ga), corrected = lef(-1),
             frees_table = lef(+1)), 5)
```

```
##                  [,1]     [,2]     [,3]
## dgamma       0.0768350 0.215100 0.067967
## corrected    0.0768350 0.215100 0.067967
## frees_table 0.0007868 0.002203 0.000696
```

Exercise 8.8 (FREES exercise 13.2)**.** For the `AutoCollision` data, fit a gamma regression with logarithmic link and claim counts as weights on age group and vehicle use, and verify FREES Table 13.4.

Solution. Example 8.12: the fitted cell means $\hat{\mu}_{ij} = e^{b_0} e^{\hat{\alpha}_i} e^{\hat{\tau}_j}$ (Proposition 8.10(iv), which needs only the logarithmic link) reproduce every entry of Table 13.4, whose first two column labels are interchanged. With claim counts as weights, a cell average of w_i independent claims of variance $\phi\mu_i^2$ has variance $\phi\mu_i^2/w_i$, the weighting of Definition 8.34. ■

Exercise 8.9 (FREES exercise 13.3)**.** Verify that the Tweedie distribution is a linear exponential family by giving $S(y, \phi)$, and verify the moments $\mathsf{E}\, T = \mu$ and $\operatorname{var} T = \phi\mu^p$, T as in Theorem 8.50.

Solution. Theorem 8.50 and (18). The series against `tweedie::dtweedie` at $\mu = 4$, $\phi = 2.5$, $p = 1.6$, and the moments by simulation:

```
p0 <- 1.6; mu0 <- 4; phi0 <- 2.5; al <- (2 - p0) / (p0 - 1)
th <- mu0^(1 - p0) / (1 - p0); bt <- mu0^(2 - p0) / (2 - p0)
S_tw <- function(u) log(sum(exp((1:300) * log(u^al / ((p0 - 1)^al * phi0^(1 + al) *
  (2 - p0))) - lgamma(2:301) - lgamma((1:300) * al) - log(u))))
```

```r
dens <- sapply(yv, function(u) exp((u * th - bt) / phi0 + S_tw(u)))
signif(rbind(series = dens, dtweedie = dtweedie(yv, xi = p0, mu = mu0, phi = phi0)),
 ↪  7)
```

```
##           [,1]   [,2]    [,3]
## series   0.153 0.1021 0.04905
## dtweedie 0.153 0.1021 0.04905
```

```r
set.seed(11); lam <- bt / phi0; gam <- 1 / (phi0 * (p0 - 1) * mu0^(p0 - 1))
Ss <- vapply(rpois(2e5, lam), function(k) if (k == 0) 0 else
  sum(rgamma(k, shape = al, rate = gam)), numeric(1))
round(c(mean = mean(Ss), theory = mu0, var = var(Ss), theory_var = phi0 * mu0^p0,
        P0 = mean(Ss == 0), theory_P0 = exp(-lam)), 3)
```

```
##       mean     theory        var theory_var         P0
##      4.005      4.000     23.005     22.974      0.175
##  theory_P0
##      0.175
```

Exercise 8.10 (SOA sample question 7)**.** Which pair of distribution and link function is most appropriate to model whether a person is hospitalised? (A) normal, identity; (B) normal, logit; (C) binomial, linear; (D) binomial, logit; (E) it cannot be determined from the information given.

Solution. The response is binary, so its law is Bernoulli, the binomial with $m = 1$ (Unit 7; Proposition 8.35); the normal (A, B) puts mass on every interval and cannot be the law of a $\{0, 1\}$ variable. The mean is a probability, and of the two links only the logit maps $(0, 1)$ onto $\mathbb{R}$, so that every $\mathbf{x}'\boldsymbol{\beta}$ gives a mean in $(0, 1)$; a linear link does not (Proposition 8.8(i)'s argument, as in Unit 7's linear probability model). The answer is (D). ■

Exercise 8.11 (SOA sample question 20)**.** A simple linear model for a probability gives predicted values below 0 and above 1. Which is the most appropriate remedy? (A) Limit the data to observations expected to give predictions in $[0, 1]$. (B) Treat predictions below 0 as 0 and above 1 as 1. (C) Use a logit function to transform the linear model into only predicting values between 0 and 1. (D) Use the canonical link of the Poisson distribution to transform the linear model into only predicting values between 0 and 1. (E) None of the above.

Solution. (A) selects observations by their covariates and still leaves a linear mean that leaves $[0,1]$ for other covariate values (Unit 7, the first drawback); (B) changes the fitted values without changing the model, which still implies negative variances $\hat{y}(1-\hat{y})$ outside $[0,1]$. (C) is the logistic model: the inverse logit maps $\mathbb{R}$ onto $(0,1)$. (D) The Poisson's canonical link is $\ln\mu$ (Proposition 8.35), whose inverse e^{η} maps $\mathbb{R}$ onto $(0,\infty)$, not into $(0,1)$. The answer is (C). ■

Exercise 8.12 (SOA sample question 45)**.** A generalised linear model with logarithmic link for an exam score on a 0–10 scale has intercept -0.1, study time (in units of 100 hours) 0.5, first attempt 0.5, master's degree -0.1, and attempt $\times$ master's interaction 0.2. An applicant with a master's degree, for whom the exam would be a first attempt, is to be offered half of the study time that gives an expected score of 6.0. Calculate the study time offered: (A) 123 hours; (B) 126; (C) 129; (D) 132; (E) 135.

Solution. With study time s (in hundreds of hours), first attempt 1, master's 1 and the interaction 1, the linear predictor is $-0.1+0.5s+0.5-0.1+0.2=0.5+0.5s$, and the logarithmic link gives $\mu=e^{0.5+0.5s}$. Setting $\mu=6$, $0.5s=\ln 6-0.5=1.29176$, so $s=2.5835$, that is 258.35

hours, and half of it is 129.2 hours: (C). ■

Exercise 8.13 (SOA sample question 52)**.** Which statements about Pearson residuals are true? I. They can be used to calculate a goodness-of-fit statistic. II. They can be used to detect if additional variables of interest can improve the model specification. III. They can be used to identify unusual observations. (A) I only; (B) II only; (C) III only; (D) I, II and III; (E) none of (A)–(D).

Solution. I: $\sum_i r_i^2$ is the Pearson statistic (Definition 8.16), whose ratio to $n-(k+1)$ tends to 1 under the model and to ϕ under (6) (Proposition 8.17). II and III: at the true means each r_i has mean 0 and variance ϕ (Proposition 8.48), the same for every observation, so a residual far from 0 marks an unusual observation, and a trend of the r_i against a candidate covariate is structure the mean has missed; the raw residuals, of variance $\phi v(\mu_i)/w_i$, mix such a trend with the variance function. All three are true: (D). ■

Exercise 8.14 (SOA sample question 75)**.** Which statement about the Tweedie distribution with $1<p<2$ is NOT true? (A) Its variance is $\phi\mu^p$. (B) The probability mass at zero is $e^{-\lambda}$. (C) It is a member of the linear exponential family. (D) The Poisson distribution is a special case of this

Tweedie distribution. (E) This Tweedie distribution is a Poisson sum of gamma random variables.

Solution. (A), (B), (C) and (E) are Theorem 8.50. For (D): with $1 < p < 2$, $\mathsf{P}(T > 0) = 1 - e^{-\lambda} > 0$ and the law of T on $(0, \infty)$ has a density, so $\mathsf{P}(T \in A) = 0$ for the countable set $A = \{1, 2, \ldots\}$ while $\mathsf{P}(T > 0) > 0$; a Poisson variable puts all its positive mass on A. So no member with $1 < p < 2$ is a Poisson, and (D) is the false statement.

As $p \downarrow 1$ with μ, ϕ fixed the law tends to that of ϕN with $N \sim \text{Pois}(\mu/\phi)$: in (17), $\lambda \to \mu/\phi$, and each Y_j has mean $\alpha/\gamma = \phi(2-p)\mu^{p-1} \to \phi$ and variance $(\alpha/\gamma)^2/\alpha \to 0$, so $\mathsf{E}\,|Y_j - \phi| \leqslant \{\text{var}\, Y_j + (\mathsf{E}\, Y_j - \phi)^2\}^{1/2} \to 0$. Conditioning on N, $\mathsf{E}\,|T - \phi N| \leqslant \mathsf{E}\, N \cdot \mathsf{E}\,|Y_1 - \phi| = \lambda\, \mathsf{E}\,|Y_1 - \phi| \to 0$, while $\phi N \xrightarrow{d} \phi\, \text{Pois}(\mu/\phi)$ because the Poisson probabilities are continuous in λ; Slutsky's theorem gives the limit. It is a Poisson only when $\phi = 1$, a limit and not a member. ■